

Local linearization based subvector inference in moment inequality models

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Abstract

This paper introduces a bootstrap-based profiling inference method for subvectors in moment inequality models following insights from [Bugni et al. \(2017\)](#). Compared to their paper, the new method calculates the critical value by searching over a local neighborhood of a pre-estimator, instead of the whole null parameter space, to profile out nuisance parameters. In this way, non-linear moment conditions are simplified by linear expansion and the bootstrap iterates over quadratic programming problems, which significantly simplifies and accelerates computation. This method controls asymptotic size uniformly over a large class of data generating processes. In the Monte Carlo simulations, the new procedure improves upon the computing time of [Bugni et al. \(2017\)](#) and [Kaïdo et al. \(2019\)](#) significantly. I provide an empirical illustration estimating an airline entry game.

KEYWORDS: Partial identification, subvector inference, moment inequalities, hypothesis testing

JEL classification: C01, C12, C15

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1 Introduction

A moment inequality model has the form

$$\begin{aligned} E_F[m_j(W_i; \theta_0)] &\geq 0, \quad j = 1, \dots, p \\ E_F[m_j(W_i; \theta_0)] &= 0, \quad j = p + 1, \dots, k \end{aligned} \tag{1}$$

where $\{W_i\}_{i=1}^n$ is an independent and identically distributed (i.i.d.) sequence of random variables with distribution F and $m = (m_1, \dots, m_k)' : \mathbb{R}^d \times \Theta \rightarrow \mathbb{R}^k$ is a known measurable function of a finite-dimensional parameter vector $\theta \in \Theta$. Frequently the parameter θ_0 in this model is not known to be point identified, a feature that complicates both estimation and inference substantially. Most of the recent papers on inference for moment inequalities focus on confidence sets for the entire parameter vector, e.g. [Andrews and Shi \(2013\)](#), [Andrews and Soares \(2010\)](#), [Chernozhukov et al. \(2007\)](#), and [Rosen \(2008\)](#). See [Canay and Shaikh \(2017\)](#) and [Molinari \(2020\)](#) for more references. However, applied researchers are often interested in a few specific components (a subvector) of the whole parameter vector, i.e. $c'\theta_0$ where c is a $d_\theta \times d_r$ matrix with full column rank.

[Bugni et al. \(2017\)](#) (BCS17) propose a bootstrap-based test procedure for $H_0 : c'\theta_0 = r$. Similar to [Romano and Shaikh \(2008\)](#), they calculate the profiled test statistic, as well as its bootstrap approximation, by minimizing the test statistic over the *whole* null parameter space to profile out the nuisance parameters. Building on their work, this paper uses the same test statistic and improves upon BCS17 by applying *local* linearization methods to calculate the critical value. The challenge of local linearization is that the a naive bootstrapped minimizer $\hat{\theta}_{n,r}^*$ may be near a different point of the identified set from the original minimizer $\hat{\theta}_{n,r}$. To address this issue, a penalty term is added to the bootstrapped test statistic to keep $\hat{\theta}_{n,r}^*$ close to $\hat{\theta}_{n,r}$, and therefore, close to the same point in the identified set. This enables linearization of the moment conditions and the constraints around this point. In this way, the bootstrap iterates over quadratic programming as opposed to general nonlinear problems. The local linearization based test (Test LL) controls asymptotic size uniformly over a large class of data generating processes and can be inverted to construct the confidence interval. As for power properties, Test LL weakly dominates projection-based procedures in any finite sample, and strictly dominates projection-based procedures in terms of asymptotic power.

[Kaido et al. \(2019\)](#) (KMS19) introduce a calibrated projection-based subvector inference method, where they calibrate the critical value so that the coverage of the subvector is above the prespecified probability. KMS19 also linearize constraints locally around the evaluated points, and the critical value can be calculated by analyzing linear programs. However, KMS conducts the test for a grid of θ ; while LL, as a profile based method, conducts the test for a grid of r only. The grid search over a relatively large dimensional space can be time-consuming, which makes the LL method more computationally efficient, especially with d_θ much larger than d_r . [Cox and Shi \(2019\)](#) propose a simple test for moment inequalities based on the rank of the active inequalities in finite samples, but it is required that the nuisance parameters enter the moment conditions linearly to adapt it for subvector inference. [Belloni et al. \(2018\)](#) study subvector inference with many moment inequalities. [Chen et al. \(2018\)](#) provide Monte Carlo

confidence sets for the identified sets, rather than the true value, of both the full parameter vector and subvectors. [Kline and Tamer \(2016\)](#) develop a Bayesian approach that can be used for subvector inference in partially identified models, but they do not provide uniform validity results.

The Monte Carlo simulation studies the finite sample properties of Test LL and compares it to BCS17 and KMS19, with all three methods implemented with the E-A-M algorithm proposed in KMS19. The simulation study, conducted on a simple entry game, considers two sample sizes with $n = 1000, 4000$ and three model sizes with the number of moment conditions $k = 8, 16, 24$ and the number of parameters $d_\theta = 7, 11, 15$. Test LL improves upon the computing time of BCS17 (resp. KMS19) by a factor ranging from 30 to 170 (resp. 3 to 27). The computational advantage of Test LL is more prominent with a larger sample and model size, reaching the highest factors for the highest dimensional parameters. The simulation results also confirm the good size and power properties.

The structure of the paper is as follows. In Section 2, I illustrate the main idea of local linearization based subvector inference. In Section 3, I compare it with BCS17 in different aspects. And then, Section 4 gives a step by step implementation guide. Section 5 reports Monte Carlo simulations comparing the finite sample properties of Test LL with that of BCS17 and KMS19. Section 6 revisits the empirical application of the airline entry game in [Kline and Tamer \(2016\)](#). Section 7 concludes the paper. Proofs are provided in the appendix.

2 Local linearization based subvector inference

In this section, I set up the model and illustrate the local linearization based test.

2.1 The model and notation

Throughout the paper, I assume that the distribution F belongs to a baseline distribution space $\mathcal{P}$, and I maintain the following mild assumption on Θ .

Assumption 1. *The parameter space Θ is a non-empty, convex and compact subset of $\mathbb{R}^{d_\theta}$ with $d_\theta < \infty$.*

Definition 1. For any distribution $F \in \mathcal{P}$, the corresponding identified set $\Theta_I(F)$ is given by

$$\Theta_I(F) := \left\{ \theta \in \Theta : \begin{array}{ll} E_F[m_j(W_i; \theta)] \geq 0, & j = 1, \dots, p \\ E_F[m_j(W_i; \theta)] = 0, & j = p + 1, \dots, k \end{array} \right\}.$$

By definition, the identified set $\Theta_I(F)$ is the set of parameter values $\theta \in \Theta$ that are observationally equivalent to the true value θ_0 . The objective of this paper is to construct a confidence interval for $c'\theta_0$ where c is a $d_\theta \times d_r$ matrix. I do this by inverting a test of the hypotheses

$$H_0 : c'\theta_0 = r \quad \text{v.s.} \quad H_1 : c'\theta_0 \neq r, \quad (2)$$

where r is a $d_r \times 1$ vector. The hypothesis test is based on a profiled test statistic $J_n(r)$ such

that H_0 is rejected if $J_n(r)$ is too large. Specifically, it takes the form

$$\phi_n^{\text{LL}}(r) := \mathbf{1} [J_n(r) > cv_n^{\text{LL}}(r, 1 - \alpha)],$$

where $\mathbf{1}[\cdot]$ is the indicator function, $\alpha \in (0, 1)$ is the significance level, and $cv_n^{\text{LL}}(r, 1 - \alpha)$ is the critical value I define later.

Before presenting the test procedure, I introduce more notation. For vector a , b , denote $\|a\| := \sqrt{a'a}$ and denote $d(a, b) := \|a - b\|$ as the Euclidean distance between a and b . For point a and set B , let $d(a, B) := \inf_{b \in B} \|a - b\|$. For a scalar α , an element a and a set A , let $a + A = \{a + \tilde{a} : \tilde{a} \in A\}$ and $\alpha A = \{\alpha a : a \in A\}$. The ε -expansion of $\Theta_1 \subseteq \Theta$ is defined as

$$\Theta_1^\varepsilon := \{\theta \in \Theta : d(\theta, \Theta_1) \leq \varepsilon\}.$$

Let

$$\Theta(r) := \{\theta \in \Theta : c'\theta = r\}$$

be the set of parameters that satisfy the subvector restriction, and

$$\Theta_I(r, F) := \{\theta \in \Theta_I(F) : c'\theta = r\}. \quad (3)$$

be the set of parameters that satisfy both the moment conditions and the subvector restriction. $\Theta_I(r, F)$ is compact and non-empty under H_0 . Define the projection operator $T^r : \Theta \rightarrow \Theta$ that projects θ on $\Theta_I(r, F)$, i.e.

$$T^r \theta \in \arg \min_{\tilde{\theta} \in \Theta_I(r, F)} d(\tilde{\theta}, \theta),$$

where $T^r \theta$ is well defined under H_0 . If the arg min set contains more than one element, $T^r \theta$ is one of the minimizers, and the choice does not affect the results. The operator T^r depends on F , but I leave this dependence implicit to simplify notation. Denote by

$$\begin{aligned} \Sigma_F(\theta) &= E_F \left[\left(m(W, \theta) - E_F[m(W, \theta)] \right) \left(m(W, \theta) - E_F[m(W, \theta)] \right)' \right] \\ D_F(\theta) &= \text{diag}(\Sigma_F(\theta)) \end{aligned}$$

the variance-covariance matrix and the diagonal matrix of variance. Let $[x]_+ = \max(x, 0)$, $[x]_- = \max(-x, 0)$ and $\mathbb{R}_+ = [0, +\infty)$.

Definition 2. The parameter space of (r, F) is defined by

$$\mathcal{L} := \{(r, F) : F \in \mathcal{P}, r \in \{c'\theta : \theta \in \Theta\}\},$$

and the null parameter space of (r, F) is

$$\mathcal{L}_0 := \{(r, F) : F \in \mathcal{P}, r \in \{c'\theta : \theta \in \Theta_I(F)\}\}.$$

2.2 The profiled test statistic

Denote by

$$\begin{aligned}\bar{m}_n(\theta) &:= \frac{1}{n} \sum_{i=1}^n m(W_i, \theta), \\ \hat{\Sigma}_n(\theta) &:= \frac{1}{n} \sum_{i=1}^n (m(W_i, \theta) - \bar{m}_n(\theta)) (m(W_i, \theta) - \bar{m}_n(\theta))',\end{aligned}$$

the sample mean and sample variance-covariance matrix of the moment functions in (1). Let

$$\hat{D}_n(\theta) := \text{diag}(\hat{\Sigma}_n(\theta))$$

be the diagonal matrix of sample variances, and

$$\hat{\Omega}_n(\theta) := \hat{D}_n(\theta)^{-\frac{1}{2}} \hat{\Sigma}_n(\theta) \hat{D}_n(\theta)^{-\frac{1}{2}}$$

be the sample correlation matrix.

Define the profiled test statistic by

$$J_n(r) := \inf_{\theta \in \Theta(r)} S(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta)) \quad (4)$$

where S is a test function specified by the econometrician and required to satisfy several regularity assumptions, see Assumptions 7, 8 in the Appendix. These assumptions are standard in the literature, see, e.g. Andrews and Guggenberger (2009) and Andrews and Soares (2010). Two widely used test functions are the modified method of moments (MMM) and the quasi-likelihood ratio (QLR) statistics given by

$$\begin{aligned}S^{\text{MMM}}(m, \Sigma) &= \sum_{j=1}^p \left[m_j / \Sigma_{jj}^{\frac{1}{2}} \right]_-^2 + \sum_{j=p+1}^k \left(m_j / \Sigma_{jj}^{\frac{1}{2}} \right)^2, \\ S^{\text{QLR}}(m, \Sigma) &= \inf_{t=(t_1, 0_{k-p}): t_1 \in \mathbb{R}_+^p} (m - t)' \Sigma^{-1} (m - t).\end{aligned}$$

The test statistic takes the infimum of test function $S(\cdot)$ over the restricted parameter space $\Theta(r)$.

2.3 Local linearization based critical value

The critical value $cv_n^{\text{LL}}(r, 1 - \alpha)$ is the $1 - \alpha$ quantile of

$$J_n^{\text{LL}}(r) := \inf_{\hat{\theta}_{n,r} \in \hat{\Theta}_I(r)} S \left(v_n^*(\hat{\theta}_{n,r}) + \varphi \left(\xi_n(\hat{\theta}_{n,r}) \right) + \hat{G}_n(\hat{\theta}_{n,r}) \Delta(\hat{\theta}_{n,r}), \hat{\Omega}_n(\hat{\theta}_{n,r}) \right). \quad (5)$$

where

$$\Delta(\hat{\theta}_{n,r}) \in \underset{\Delta \in \mathbf{\Delta}_n^r(\hat{\theta}_{n,r})}{\text{argmin}} \left(S \left(v_n^*(\hat{\theta}_{n,r}) + \varphi \left(\xi_n(\hat{\theta}_{n,r}) \right) + \hat{G}_n(\hat{\theta}_{n,r}) \Delta, \hat{\Omega}_n(\hat{\theta}_{n,r}) \right) + \frac{\lambda_n}{n} \|\Delta\|^2 \right). \quad (6)$$

The elements in (5) and (6) are listed below.

1. $v_n^*(\cdot)$ is a stochastic process defined as

$$v_n^*(\theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{D}_n(\theta)^{-\frac{1}{2}} (m(W_i, \theta) - \bar{m}_n(\theta)) \zeta_i, \quad \zeta_i | \{W_i\}_{i=1}^n \sim N(0, 1) \text{ i.i.d.} \quad (7)$$

2. $\xi_n(\theta) = \frac{\sqrt{n}}{\kappa_n} \hat{D}_n(\theta)^{-\frac{1}{2}} \bar{m}_n(\theta)$ is a slackness function with $\kappa_n \rightarrow \infty$ and $\kappa_n = O\left(\frac{n^{1/4}}{(\ln n)^{1/\chi}}\right)$ where χ is defined in Assumption 8.5. Following Andrews and Soares (2010), the suggested choice is $\kappa_n = \sqrt{\ln n}$.

3. $\varphi(\cdot) = (\varphi_1(\cdot), \dots, \varphi_k(\cdot))$ is a generalized moment selection (GMS) function with

$$\varphi_j(\xi) = \begin{cases} [\xi_j]_+ & \text{for } j = 1, \dots, p \\ 0 & \text{for } j = p + 1, \dots, k. \end{cases} \quad (8)$$

4. $\hat{G}_n(\cdot)$ is a consistent estimator of the Jacobian $G_F(\theta) = \nabla_{\theta'} \left(D_F(\theta)^{-\frac{1}{2}} E_F(m(\theta)) \right)$.
5. λ_n is a non-random sequence satisfying $\lambda_n/\sqrt{n} \rightarrow \infty$ and $\lambda_n (\ln \kappa_n^2)^{\frac{2}{\chi}} = O(n)$ where χ is defined in Assumption 8.5. A valid choice is $\lambda_n = \sqrt{n} \kappa_n^2$.
6. $\hat{\Theta}_I(r)$ is a non-empty subset of the minimizers

$$\emptyset \neq \hat{\Theta}_I(r) \subseteq \underset{\theta \in \Theta(r)}{\operatorname{argmin}} S \left(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta) \right). \quad (9)$$

More generally, $\hat{\Theta}_I(r)$ can in fact be a subset of approximate minimizers satisfying

$$\sup_{\theta \in \hat{\Theta}_I(r)} d(\theta, \Theta_I(r, F)) = \sup_{\theta \in \hat{\Theta}_I(r)} d(\theta, T^r \theta) = o_p \left(n^{-1/2} (\ln \kappa_n^2)^{1/\chi} \right) \quad (10)$$

with χ defined in Assumption 8.5.

7. $\Delta_n^r(\cdot)$ is a set such that $\Delta_n^r(\theta) \subseteq \sqrt{n}(\Theta(r) - T^r \theta)$ for all $\theta \in \hat{\Theta}_I(r)$ with probability approaching one.

To illustrate the construction of the critical value, first, omit the outer inf operator in (5) and focus on a single $\hat{\theta}_{n,r} \in \hat{\Theta}_I(r)$. The first step estimator, $\hat{\theta}_{n,r}$, needs to be recalculated for each bootstrap draw. The problem with doing a naive bootstrap is that it may be near a different point of the identified set rather than $T^r \hat{\theta}_{n,r}$. A modified bootstrapped version of $\hat{\theta}_{n,r}$ is given by $T^r \hat{\theta}_{n,r} + \frac{1}{\sqrt{n}} \Delta(\hat{\theta}_{n,r})$ with $\Delta(\hat{\theta}_{n,r})$ defined in (6). In that expression, the first element $S(\cdot)$ is a local linearization of the test function around $T^r \hat{\theta}_{n,r}$.¹ The second element is a penalty term that keeps the bootstrap version of $\hat{\theta}_{n,r}$ near $T^r \hat{\theta}_{n,r}$, the same point in the identified set. The feasible set of $T^r \hat{\theta}_{n,r} + \frac{1}{\sqrt{n}} \Delta(\hat{\theta}_{n,r})$ is $\Theta(r)$, hence we need to choose $\Delta(\hat{\theta}_{n,r}) \in \sqrt{n} \left(\Theta(r) - T^r \hat{\theta}_{n,r} \right)$. However, $T^r \hat{\theta}_{n,r}$ is unknown and thus we approximate the boundary constraint by $\Delta_n^r(\hat{\theta}_{n,r})$.

¹In (6), we can replace $T^r \hat{\theta}_{n,r}$ with $\hat{\theta}_{n,r}$ since they are close enough.

When the bootstrapped minimizer is plugged into the linearized statistic, the result is (5). Lastly, note that this local optimization holds for all $\hat{\theta}_{n,r} \in \hat{\Theta}_I(r)$, thus we implement the outer inf operator in (5) to improve power.

The penalty term in (6) is a new contribution of this paper, and it validates the local linearization approximation of the moment conditions, which greatly improves the computation. When Test LL is implemented with the MMM or QLR test function and a polyhedral $\Delta_n^r(\theta)$, the inner minimization in (6) is a simple quadratic programming problem. Also, note that $\hat{\Theta}_I(r)$ is not required to contain all the minimizers for the coverage property, and in practice, $\hat{\Theta}_I(r)$ contains only a few elements. In this way, the bootstrap iterates over quadratic programming problems, which can be solved with high-speed solvers like CVXGEN in [Mattingley and Boyd \(2012\)](#).

Next, I present the assumptions that are important for proper coverage.

Assumption 2. For all $(r, F) \in \mathcal{L}_0$ and $\theta \in \Theta(r)$,

$$S\left(E_F(m(W, \theta)), \Sigma_F(\theta)\right) \geq c \min\{\delta, d(\theta, \Theta_I(r, F))\}^\chi$$

for some constants $c, \delta > 0$ and for χ as in Assumption 8.5.

Assumption 2 is borrowed from BCS17. It states that in a neighborhood of the null identified set $\Theta_I(r, F)$, the population version of the test function can be bounded below by a polynomial of the deviation of θ from $\Theta_I(r, F)$. It is similar to the polynomial minorant condition in [Chernozhukov et al. \(2007\)](#) (Eqs. (4.1) and (4.5)), but logically independent. This assumption rules out some cases where the entire boundary is a smooth manifold, focusing on a simpler class of models with a well-behaved $\Theta_I(r, F)$.² However, it holds under many cases of practical interest. [Kaido et al. \(2022\)](#) extensively analyze the constraint qualifications in partial identification, including the minorant type condition, and so I refer to their paper for detailed discussion. Assumption 2, together with other regularity assumptions, guarantees that $\hat{\theta}_{n,r}$ is close enough to $T^r \hat{\theta}_{n,r}$, so that we can replace $T^r \hat{\theta}_{n,r}$ with $\hat{\theta}_{n,r}$ in the linear approximation.

Assumption 3. For all distributions $F \in \mathcal{P}$, it holds that

1. For each j , there exist $G_{F,j}(\theta) = \nabla_{\theta'} \left(D_{F,j}(\theta)^{-\frac{1}{2}} E_{F,j}(m(W, \theta)) \right)$ and its estimator $\hat{G}_{n,j}(\cdot)$ such that

$$\sup_{\theta \in \Theta} \left\| \hat{G}_{n,j}(\theta) - G_{F,j}(\theta) \right\| = O_p \left(\frac{\kappa_n}{\sqrt{n}} \right)$$

uniformly in $\mathcal{P}$, where κ_n is the GMS tuning parameter.

2. There exists $M_G, \bar{M}_G < \infty$, such that for all $\theta_1, \theta_2 \in \Theta$,

$$\max_{j=1, \dots, k} \sup_{F \in \mathcal{P}} \|G_{F,j}(\theta_1) - G_{F,j}(\theta_2)\| \leq M_G \|\theta_1 - \theta_2\|$$

²To give a simple example excluded by Assumption 2, let $E[W] = 1$, $\text{var}(W) = 1$, $\Theta = [-2, 2]^2$, $m(W; \theta) = E[W - \theta_1^2 - \theta_2^2] \geq 0$ and $H_0 : \theta_1 = 1$. It is easy to get $\Theta_I(r, F) = \{(1, 0)\}$. Consider the MMM test statistic. For all $\theta_2 > 0$, $d((1, \theta_2), \Theta_I(r, F)) = \theta_2$ and $S\left(E_F(m(W_i, \theta)), \Sigma_F(\theta)\right) = [-\theta_2^2]^2 = \theta_2^4$. In this case, $\chi = 2$ and we can not find c, δ such that Assumption 2 holds.

and

$$\max_{j=1,\dots,k} \sup_{F \in \mathcal{P}} \sup_{\theta \in \Theta_I(F)} \|G_{F,j}(\theta)\| \leq \bar{M}_G.$$

Assumption 3 requires that the normalized population moments are differentiable and that the derivative can be consistently estimated uniformly in θ and in F . For instance, in a separable model with $D_F(\theta) = D_F$ invariant to θ , we can use

$$\hat{G}_{n,j}(\theta) = \hat{D}_{n,j}^{-\frac{1}{2}} \nabla_{\theta'} \bar{m}_j(\theta).$$

See KMS19 Supplemental Appendix F for such estimators in some canonical moment inequality examples.

Lastly, I give a sufficient primitive assumption for the set $\Delta_n^r(\theta)$.

Assumption 4. Set $\Delta_n^r(\theta)$ satisfies:

1. $0 \in \Delta_n^r(\theta)$.
2. Let $\varepsilon_n = n^{-1/2}(\ln \kappa_n^2)^{1/\chi}$ where χ satisfies Assumption 8.5. There exists $a > 0$ such that

$$\liminf_n \inf_{(r,F) \in \mathcal{L}_0} \inf_{\theta \in \Theta(r) \cap \Theta_I(r,F)^{a\varepsilon_n}} P(\Delta_n^r(\theta) \subseteq \sqrt{n}(\Theta(r) - T^r\theta)) = 1. \quad (11)$$

$\Delta_n^r(\theta)$ is a key choice for implementing the method and sets the boundary for the local search to get the critical value. Note that $\Theta_I(r, F)^{a\varepsilon_n}$ is a non-random expansion of $\Theta_I(r, F)$ and

$$\hat{\theta}_{n,r} \in \hat{\Theta}_I(r) \subseteq \Theta(r) \cap \Theta_I(r, F)^{a\varepsilon_n}$$

with probability approaching one. Thus Assumption 4 guarantees that

$$\Delta_n^r(\hat{\theta}_{n,r}) \subseteq \sqrt{n}(\Theta(r) - T^r\hat{\theta}_{n,r})$$

holds asymptotically. Assumption 4 is trivially satisfied with $\Delta_n^r(\theta) = \{0_{d_\theta}\}$, but the idea is to choose $\Delta_n^r(\theta)$ as large as possible, while still satisfying Assumption 4, to improve the power. To give a simple example of $\Delta_n^r(\theta)$, let $\Theta = [0, 1]^2$, $c = (1, 0)$ and $\Theta(r) = \{r\} \times [0, 1]$. Note that for $\theta = (\theta_1, \theta_2) \in \Theta(r) \cap \Theta_I(r, F)^{\varepsilon_n}$, $d(\theta, T^r\theta) \leq \varepsilon_n$, thus

$$\begin{aligned} \sqrt{n}(\Theta(r) - T^r\theta) &= \sqrt{n}(\Theta(r) - \theta) + \sqrt{n}(\theta - T^r\theta) \\ &= \{0\} \times [\sqrt{n}(0 - \theta_2) + \sqrt{n}(\theta_2 - (T^r\theta)_2), \sqrt{n}(1 - \theta_2) + \sqrt{n}(\theta_2 - (T^r\theta)_2)] \\ &\supseteq \{0\} \times [\sqrt{n}(-\theta_2 + \varepsilon_n), \sqrt{n}(1 - \theta_2 - \varepsilon_n)] \end{aligned}$$

where the first two lines are straightforward and the last line follows from $|\theta_2 - (T^r\theta)_2| \leq d(\theta, T^r\theta) \leq \varepsilon_n$. Hence,

$$\Delta_n^r(\theta) = (\{0\} \times [\sqrt{n}(-\theta_2 + \varepsilon_n), \sqrt{n}(1 - \theta_2 - \varepsilon_n)]) \cup \{0_2\} \quad (12)$$

satisfies Assumption 4. Also, note that Assumption 4 allows for data-dependent choices of tuning parameters that enter $\Delta_n^r(\theta)$ since $\Delta_n^r(\theta) \subseteq \sqrt{n}(\Theta(r) - T^r\theta)$ only needs to hold with

probability approaching one. Section 4.1 gives more examples.

Theorem 1. *Let Assumptions 1, 2, 3, 4, 6, 7, 8, 9 hold. Then for all $\alpha \in (0, \frac{1}{2})$,*

$$\limsup_{n \rightarrow \infty} \sup_{(r, F) \in \mathcal{L}_0} E_F [\phi_n^{\text{LL}}(r)] \leq \alpha. \quad (13)$$

Assumptions 6, 7, 8, 9 can be found in Appendix A.2. Assumptions 6 and 7 are borrowed from BCS17. Assumption 6 gives regularity conditions on the baseline distribution space. It is the same as BCS17 Definition 4.2 and thus I refer to their paper for detailed discussion. Assumption 7 is a continuity assumption on the limit distribution of the test statistic. Assumption 8 is the maintained assumption on the test function S . Assumption 9 restricts the rates of tuning parameters.

Remark 1. The set $\hat{\Theta}_I(r)$ is also used in BCS17 with a similar requirement, and BCS17 Remark 4.1 applies here. The key restriction for set $\hat{\Theta}_I(r)$ is that

$$\sup_{\theta \in \hat{\Theta}_I(r)} d(\theta, \Theta_I(r, F)) = o_p \left(n^{-1/2} (\ln \kappa_n^2)^{1/\chi} \right)$$

uniformly, or sufficiently

$$P \left(\hat{\Theta}_I(r) \subseteq \Theta(r) \cap \Theta_I(r, F)^{\varepsilon_n} \right) \rightarrow 1$$

uniformly with $\varepsilon_n = an^{-1/2} (\ln \kappa_n^2)^{1/\chi}$ for some $a > 0$. This has two implications. First, we do not need to consistently estimate $\Theta_I(r, F)$ and $\hat{\Theta}_I(r)$ is not required to contain all the minimizers of $J_n(r)$. The intuition is that with fewer elements in $\hat{\Theta}_I(r)$, the critical value defined in (5) is larger and the coverage is higher. And thus for the asymptotic size control, it suffices to include only one minimizer and construct the critical value by minimizing locally around the single point. Second, $\hat{\Theta}_I(r)$ can also be a subset of approximate minimizers defined as

$$\hat{\Theta}_I(r) \subseteq \left\{ \theta \in \Theta(r) : S(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta)) \leq J_n(r) + \delta_n \right\}$$

with $\delta_n \geq 0$ and $\delta_n = o_p(1)$. I give detailed guidance on calculating $\hat{\Theta}_I(r)$ in Section 4.2.

3 Comparison with BCS critical value

BCS17 use the same profiled test statistic as in (4). In this section, I compare the LL critical value with BCS critical value in terms of the choice of GMS functions, power properties and computation.

3.1 BCS critical value

Let $v_n^*(\cdot)$ defined as in (7). And to simplify notation, let

$$Q_n \left(\tilde{\Theta}, \tilde{\varphi}(\cdot), \tilde{\Delta}(\cdot) \right) = \inf_{\theta \in \tilde{\Theta}} S \left(v_n^*(\theta) + \tilde{\varphi}(\theta) + \hat{G}_n(\theta) \Delta(\theta), \hat{\Omega}_n(\theta) \right) \quad (14)$$

where

$$\Delta(\theta) \in \arg \min_{\Delta \in \hat{\Delta}(\theta)} S \left(v_n^*(\theta) + \tilde{\varphi}(\theta) + \hat{G}_n(\theta)\Delta(\theta), \hat{\Omega}_n(\theta) \right) + \frac{\lambda_n}{n} \|\Delta\|^2.$$

The LL critical value is given by the $1 - \alpha$ quantile of

$$J_n^{\text{LL}}(r) = Q_n \left(\hat{\Theta}_I(r), \varphi(\xi_n(\cdot)), \mathbf{\Delta}_n^r(\cdot) \right). \quad (15)$$

The critical value of the minimum resampling test in BCS17 is given by the $1 - \alpha$ quantile of

$$J_n^{\text{BCS}}(r) = \min\{J_n^{\text{DR}}(r), J_n^{\text{PR}}(r)\}$$

with

$$J_n^{\text{DR}}(r) = Q_n \left(\hat{\Theta}_I(r), \varphi^{\text{BCS}}(\xi_n(\cdot)), \{0_{d_\theta}\} \right), \quad (16)$$

$$J_n^{\text{PR}}(r) = Q_n \left(\Theta(r), \xi_n(\cdot), \{0_{d_\theta}\} \right), \quad (17)$$

where $\varphi_n^{\text{BCS}}(\cdot)$ is a GMS function satisfying the following assumption.

Assumption 5. *Given the GMS function $\varphi^{\text{BCS}}(\cdot)$, there is a function $\varphi^* : \mathbb{R}_{[\pm\infty]}^k \rightarrow \mathbb{R}_{[+\infty]}^k$ that takes the form $\varphi^*(\xi) = (\varphi_1^*(\xi_1), \dots, \varphi_p^*(\xi_p), \mathbf{0}_{k-p})$ and, for all $j = 1, \dots, p$,*

1. $\varphi_j^*(\xi_j) \geq \varphi_j(\xi_j)$ for all $\xi_j \in \mathbb{R}_{[+\infty]}$
2. $\varphi_j^*(\cdot)$ is continuous
3. $\varphi_j^*(\xi_j) = 0$ for all $\xi_j \leq 0$ and $\varphi_j^*(\infty) = \infty$.

Assumption 5 is satisfied when φ^{BCS} is any of the functions $\varphi^{(1)} - \varphi^{(4)}$ described in [Andrews and Soares \(2010\)](#). For instance, in the main text and the simulation in BCS17, they use

$$\varphi_j^{(1)}(\xi) = \begin{cases} \infty, & \text{if } \xi_j > 1 \text{ and } j = 1, \dots, p \\ 0, & \text{if } \xi_j \leq 1, \text{ or } j = p+1, \dots, k \end{cases}. \quad (18)$$

Here and after, denote Test BCS (resp. DR, PR) as the test with the test statistic in (4) and $1 - \alpha$ quantile of $J_n^{\text{BCS}}(r)$ (resp. $J_n^{\text{DR}}(r)$, $J_n^{\text{PR}}(r)$) being the critical value.

Comparing (15), (16) and (17), we can see that J_n^{PR} minimizes over the whole null parameter space $\Theta(r)$, but J_n^{LL} and J_n^{DR} only search around $\hat{\Theta}_I(r)$. Compared to Test DR, Test LL further searches around $\hat{\Theta}_I(r)$ by minimizing over Δ , while Test DR imposes $\Delta = 0$.

3.2 GMS functions

For Test LL, I only consider $\varphi(\xi)$ defined in (8), which is $\varphi^{(3)}$ in [Andrews and Soares \(2010\)](#). Other GMS functions allowing for $\varphi_j(\xi) = \infty$ for $\xi \in \mathbb{R}^k$ may lead to overrejection. To see this, recall that we use

$$v_n^*(\hat{\theta}_{n,r}) + \varphi_n(\xi_n(\hat{\theta}_{n,r})) + \hat{G}_n(\hat{\theta}_{n,r})\Delta(\hat{\theta}_{n,r})$$

as a linear approximation to

$$D_F(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta) \approx \nu_n(T^r \hat{\theta}_{n,r}) + h_{n,F}(T^r \hat{\theta}_{n,r}) + G_F(\bar{\theta}_{n,r}) \sqrt{n}(\theta - T^r \hat{\theta}_{n,r})$$

with $\Delta(\hat{\theta}_{n,r}) = \sqrt{n}(\theta - T^r \hat{\theta}_{n,r}) = o_p(n^{\frac{1}{4}})$. Note that by construction, θ can be far away from $T^r \hat{\theta}_{n,r}$, in the sense that the j th moment inequality condition can be violated at θ even if it is slack at $T^r \hat{\theta}_{n,r}$. To be more specific, even if $h_{n,F,j}(T^r \theta_{n,r})/\kappa_n \rightarrow +\infty$, it is not necessary that

$$h_{n,F,j}(T^r \hat{\theta}_{n,r}) + G_{F,j}(\bar{\theta}_{n,r})\sqrt{n}(\theta - T^r \hat{\theta}_{n,r}) \geq 0.$$

Hence, we can not simply omit the j th moment by setting $\varphi_j = \infty$. Using a GMS function with $\varphi_j(\xi) < \infty$ for $\xi \in \mathbb{R}^k$, $j = 1, \dots, p$ introduces a penalty when violating the inequalities, which mimics the behavior of the profiled test statistic.

This also suggests that the GMS function depends closely on how the feasible set $\Delta_n^r(\theta)$ was chosen. If we choose a smaller $\Delta_n^r(\theta)$ so that it belongs to a fixed compact set, then $\Delta(\hat{\theta}_{n,r}) = O_p(1)$ and $\theta = T^r \hat{\theta}_{n,r} + \frac{1}{\sqrt{n}}\Delta(\hat{\theta}_{n,r})$ is close enough to $T^r \hat{\theta}_{n,r}$. Therefore, the j th moment inequality condition is slack at θ as long as it is large enough at $T^r \hat{\theta}_{n,r}$. That is, $h_{n,j}(T^r \theta_{n,r})/\kappa_n > 1$ implies

$$h_{n,F,j}(T^r \theta_{n,r}) + G_{F,j}(\theta_{n,r})\Delta(\theta_{n,r}) \rightarrow +\infty.$$

In this case, we can ignore the j th moment in the approximation, and $\varphi^{(1)}$ is a valid choice. Test DR can be viewed as an extreme case here with $\Delta_n^r(\theta) = \{0_{d_\theta}\}$.

In general, the power increases (weakly) in $\Delta_n^r(\theta)$ and we want it to be as large as possible. Hence, I work with $\varphi^{(3)}$ and do not impose further boundary restrictions on $\Delta_n^r(\theta)$.

3.3 Power properties

In this section, I compare Test LL with Test DR and Test PR in terms of power properties.

Comparison with Test DR

Test DR is closely related to Test LL. If we assume the GMS functions φ_n are the same in Test LL and Test DR, we have

$$\begin{aligned} J_n^{LL}(r) &= \inf_{\theta \in \hat{\Theta}_I(r)} S\left(v_n^*(\theta) + \varphi(\xi_n(\theta)) + \hat{G}_n(\theta)\Delta(\theta), \hat{\Omega}_n(\theta)\right) \\ &\leq \inf_{\theta \in \hat{\Theta}_I(r)} \left(S(v_n^*(\theta) + \varphi(\xi_n(\theta)) + \hat{G}_n(\theta)\Delta(\theta), \hat{\Omega}_n(\theta)) + \frac{\lambda_n}{n}\|\Delta(\theta)\|^2\right) \\ &\leq \inf_{\theta \in \hat{\Theta}_I(r)} S(v_n^*(\theta) + \varphi(\xi_n(\theta)), \hat{\Omega}_n(\theta)) = J_n^{\text{DR}}(r) \end{aligned}$$

where the first line is by (5), the first inequality is straightforward and the second one follows from (6) and $0 \in \Delta_n^r(\theta)$. Thus the power of Test LL is greater than Test DR for all sample sizes and all alternatives.

Remark 2. It is also common practice to use a projection-based method in applied work. The projection-based method first computes the confidence set for the entire parameter θ by

$$\text{CS}_n^\theta(1 - \alpha) \equiv \left\{ \theta \in \Theta : S(\hat{D}_n(\theta)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\theta), \hat{\Omega}_n(\theta)) \leq cv_n^\theta(\theta, 1 - \alpha) \right\}$$

where $cv_n^\theta(\theta, 1 - \alpha)$ is such that $CS_n^\theta(1 - \alpha)$ has the correct asymptotic size; and then rejects H_0 if $\Theta(r) \cap CS_n^\theta(1 - \alpha) = \emptyset$. BCS17 proves that Test DR has higher power than this projection-based test in finite samples for all alternatives. Thus, for all samples, it follows that $\phi_n^{\text{BP}}(r) \leq \phi_n^{\text{DR}}(r) \leq \phi_n^{\text{LL}}(r)$, where $\phi_n^{\text{BP}}(r)$, $\phi_n^{\text{DR}}(r)$ and $\phi_n^{\text{LL}}(r)$ be the projection-based test, Test DP in BCS17 and Test LL, respectively. By Remark 4.6 in BCS17, under a condition similar to Bugni et al. (2015)(BCS15) Assumption A.9, Test DR has *strictly* higher asymptotic power than the projection-based test. Under the same condition, Test LL has *strictly* higher asymptotic power than the projection-based test.

Example 1 illustrates a case where Test LL has strictly better asymptotic power than Test DR with MMM test function and the same GMS function φ in (8).

Example 1. Let $\{W_i\}_{i=1}^n = \{(W_{1,i}, W_{2,i}, W_{3,i})\}_{i=1}^n$ be an i.i.d. sequence of random variables with distribution $F_n, \Sigma_{F_n} = I_3, E_{F_n}[W_1] = \mu_1\kappa_n/\sqrt{n}, E_{F_n}[W_2] = \mu_2/\sqrt{n}$, and $E_{F_n}[W_3] = 0$ for some $\mu_1 > 0, \mu_2 \leq 0$. Consider the following model with $\Theta = [-1, 1]^3$:

$$\begin{aligned} E_{F_n}[m_1(W_i, \theta)] &= E_{F_n}[W_{1,i} - \theta_1] \geq 0, \\ E_{F_n}[m_2(W_i, \theta)] &= E_{F_n}[W_{2,i} - \theta_2] \geq 0, \\ E_{F_n}[m_3(W_i, \theta)] &= E_{F_n}[W_{3,i} - \theta_3] = 0. \end{aligned}$$

We are interested in testing the hypotheses

$$H_0 : c'\theta = (\theta_1, \theta_2) = (0, 0) \text{ v.s. } H_1 : c'\theta = (\theta_1, \theta_2) \neq (0, 0).$$

Below I use $Z = (Z_1, Z_2, Z_3) \sim N(0, I_3)$. Simple algebra shows that

$$\begin{aligned} J_n(r) &= \inf_{\theta_3 \in [-1, 1]} \left[\sqrt{n}\hat{\sigma}_{n,1}^{-1}\bar{W}_{n,1} \right]_-^2 + \left[\sqrt{n}\hat{\sigma}_{n,2}^{-1}\bar{W}_{n,2} \right]_-^2 + \left(\sqrt{n}\hat{\sigma}_{n,3}^{-1}(\bar{W}_{n,3} - \theta_3) \right)^2 \\ &\xrightarrow{d} [Z_2 + \mu_2]_-^2. \end{aligned}$$

Test LL Let $\hat{\Theta}_I = \{(0, 0, \bar{W}_{n,3})\}$. Let $\varepsilon_n = \sqrt{\frac{\ln \kappa_n^2}{n}}$ and

$$\Delta_n^r(\hat{\theta}_n) = \left\{ (0, 0, \tilde{\Delta}_3) : \tilde{\Delta}_3 \in \sqrt{n}[-1 + \varepsilon_n - \bar{W}_3, 1 - \varepsilon_n - \bar{W}_3] \right\} \cup \{0_3\}$$

in the spirit of (12). This test uses the (conditional) $(1 - \alpha)$ quantile of

$$J_n^{\text{LL}}(r) = \left[v_{n,1}^* + \varphi_1(\xi_n(\hat{\theta}_n)) \right]_-^2 + \left[v_{n,2}^* + \varphi_2(\xi_n(\hat{\theta}_n)) \right]_-^2 + (v_{n,3}^* - \Delta_3)^2$$

$$\Delta_3 \in \arg \min_{\tilde{\Delta} : (0, 0, \tilde{\Delta}) \in \Delta_n^r(\hat{\theta}_n)} \left[v_{n,1}^* + \varphi_1(\xi_n(\hat{\theta}_n)) \right]_-^2 + \left[v_{n,2}^* + \varphi_2(\xi_n(\hat{\theta}_n)) \right]_-^2 + (v_{n,3}^* - \tilde{\Delta})^2 + \frac{\lambda_n}{n} \tilde{\Delta}^2.$$

v_n^* is defined in (7) and it is invariant to θ in this additively separable model, thus I omit the argument. It is straightforward to get $\Delta_3 = v_{n,3}^*/(1 + \lambda_n/n)$ with probability approaching one and

$$J_n^{\text{LL}}(r) \xrightarrow{d} [Z_1 + \mu_1]_-^2 + [Z_2]_-^2.$$

Test DR We have

$$J_n^{\text{DR}}(r) = \left[v_{n,1}^* + \varphi_1(\xi_n(\hat{\theta}_n)) \right]_-^2 + \left[v_{n,2}^* + \varphi_2(\xi_n(\hat{\theta}_n)) \right]_-^2 + (v_{n,3}^*)^2 \\ \xrightarrow{d} [Z_1 + \mu_1]_-^2 + [Z_2]_-^2 + Z_3^2.$$

Thus we have $J_n^{\text{DR}}(r) > J_n^{\text{LL}}(r)$, and Test LL has strictly higher power than Test DR.

Comparison with Test PR

I first consider a simple separable linear model

$$\begin{aligned} E[m_j(W; \theta)] &= E[W_j + G_j \theta] \geq 0 & j = 1, \dots, p \\ E[m_j(W; \theta)] &= E[W_j + G_j \theta] = 0 & j = p+1, \dots, k \end{aligned}$$

where G_j is $1 \times d_\theta$ nonrandom known vector.

Test PR uses the (conditional) $(1 - \alpha)$ quantile of

$$J_n^{\text{PR}}(r) = \inf_{\theta \in \Theta(r)} S \left(v_n^* + \frac{\sqrt{n}}{\kappa_n} \hat{D}_n^{-\frac{1}{2}} (\bar{W} + G\theta), \hat{\Omega}_n \right) \quad (19)$$

$$= \inf_{\hat{\theta}_n \in \hat{\Theta}_I} \inf_{\theta \in \Theta(r)} S \left(v_n^* + \frac{\sqrt{n}}{\kappa_n} \hat{D}_n^{-\frac{1}{2}} (\bar{W} + G\hat{\theta}_{n,r}) + \frac{\sqrt{n}}{\kappa_n} \hat{D}_n^{-\frac{1}{2}} G(\theta - \hat{\theta}_{n,r}), \hat{\Omega}_n \right) \quad (20)$$

$$= \inf_{\hat{\theta}_n \in \hat{\Theta}_I(r)} S \left(v_n^* + \xi_n(\hat{\theta}_{n,r}) + \hat{D}_n^{-\frac{1}{2}} G \Delta^{\text{PR}}(\hat{\theta}_{n,r}), \hat{\Omega}_n \right) \quad (21)$$

where

$$\Delta^{\text{PR}}(\theta) \in \arg \min_{\Delta \in \frac{\sqrt{n}}{\kappa_n}(\Theta(r) - \theta)} S \left(v_n^* + \xi_n(\theta) + \hat{D}_n^{-\frac{1}{2}} G \Delta, \hat{\Omega}_n \right).$$

Note that (19) follows from the definition in (17) and (20) simply rearranges terms by adding and subtracting $\frac{\sqrt{n}}{\kappa_n} \hat{D}_n^{-\frac{1}{2}} \hat{\theta}_{n,r}$. In (21), I replace the optimal value of $\frac{\sqrt{n}}{\kappa_n}(\theta - \hat{\theta}_{n,r})$ with $\Delta^{\text{PR}}(\hat{\theta}_{n,r})$.

Recall that Test LL uses the (conditional) $(1 - \alpha)$ quantile of

$$J_n^{\text{LL}}(r) = \inf_{\hat{\theta}_n \in \hat{\Theta}_I(r)} S \left(v_n^* + \varphi(\xi_n(\hat{\theta}_{n,r})) + \hat{D}_n^{-\frac{1}{2}} G \Delta^{\text{LL}}(\hat{\theta}_{n,r}), \hat{\Omega}_n \right) \quad (22)$$

$$\Delta^{\text{LL}}(\theta) \in \arg \min_{\Delta \in \Delta_n^r(\theta)} S \left(v_n^* + \varphi(\xi_n(\theta)) + \hat{D}_n^{-\frac{1}{2}} G \Delta, \hat{\Omega}_n \right) + \frac{\lambda_n}{n} \|\Delta\|^2. \quad (23)$$

The differences between (21) and (22) are the slackness function and the choice of Δ . The two slackness functions are asymptotically equivalent under the null and local alternatives, in the sense that

$$\xi_n(\hat{\theta}_{n,r}) = \varphi_n(\xi_n(\hat{\theta}_{n,r})) + o_p(1).$$

Δ^{PR} and Δ^{LL} differ in (i) the feasible set, $\Delta^{\text{LL}}(\hat{\theta}_{n,r}) \in \Delta_n^r(\hat{\theta}_{n,r}) \subseteq \sqrt{n}(\Theta(r) - T\hat{\theta}_{n,r})$, while $\Delta^{\text{PR}} \in \frac{\sqrt{n}}{\kappa_n}(\Theta(r) - \hat{\theta}_{n,r})$; (ii) the maximum divergence rate, $\Delta^{\text{LL}} = O_p(\sqrt{\frac{n}{\lambda_n}}) = o(n^{\frac{1}{4}})$ due to the penalty term in the objective function, while $\Delta^{\text{PR}} = O_p(\frac{\sqrt{n}}{\kappa_n})$ due to the feasible set. However, if $\hat{\theta}_{n,r}$ is far away from the boundary of Θ and $\Delta^{\text{PR}}, \Delta^{\text{LL}} = O_p(1)$, the feasible set

constraint is not binding. In this case, Test LL and Test PR have the same asymptotic power under the null and the local alternatives. I use Example 1 to illustrate.

Example 1 (continued). *Test PR.*

$$J_n^{\text{PR}}(r) = \inf_{\theta_3 \in [-1, 1]} \left[v_{n,1}^* + \frac{\sqrt{n}}{\kappa_n} \hat{\sigma}_{n,1}^{-1} \bar{W}_{n,1} \right]_-^2 + \left[v_{n,2}^* + \frac{\sqrt{n}}{\kappa_n} \hat{\sigma}_{n,2}^{-1} \bar{W}_{n,2} \right]_-^2 + \left(v_{n,3}^* + \frac{\sqrt{n}}{\kappa_n} \hat{\sigma}_{n,3}^{-1} (\bar{W}_{n,3} - \theta_3) \right)^2 \\ \xrightarrow{d} [Z_1 + \mu_1]_-^2 + [Z_2]_-^2$$

with $\Delta^{\text{PR}} = \hat{\sigma}_{n,3} v_{n,3}^* = O_p(1)$. The asymptotic distribution of $J_n^{\text{PR}}(r)$ is the same as $J_n^{\text{LL}}(r)$. Note that in this example,

$$\hat{\theta}_{n,r} = (0, 0, \bar{W}_{n,3}) \xrightarrow{p} 0_3$$

is an interior point of Θ and recall that $\Delta^{\text{LL}} = (0, 0, v_{n,3}^*/(1 + \lambda_n/n)) = O_p(1)$. As expected, under the null ($\mu_2 = 0$) and local alternative ($\mu_2 < 0$), Test LL and Test PR have the same asymptotic power.

However, in a non-separable or nonlinear model, the previous comparison does not hold as the random process v_n^* , Jacobian $\hat{G}$ and correlation matrix $\hat{\Omega}$ change with θ , in which case the power comparison of Test PR and Test LL is unclear. Below I give one example where Test LL is dominated by Test PR.

Example 2. Let $\{W_i\}_{i=1}^n = \{(W_{1,i}, W_{2,i}, W_{3,i})\}_{i=1}^n$ be an i.i.d. sequence of random variables with distribution $F_n, \Sigma_{F_n} = I_3, E_{F_n}[W_1] = E_{F_n}[W_2] = 2 - \mu/\sqrt{n}, E_{F_n}[W_3] = 0$ for some $\mu \geq 0$. Consider the following model with $\Theta = [-2, 2]^2$:

$$E_{F_n}[m_1(W_i, \theta)] = E_{F_n}[W_{1,i} - \theta_1 + \theta_2] \geq 0, \\ E_{F_n}[m_2(W_i, \theta)] = E_{F_n}[W_{2,i} - \theta_1 - \theta_2] \geq 0, \\ E_{F_n}[m_3(W_i, \theta)] = E_{F_n}[W_{3,i} - \theta_1 + 1 + \eta(\theta_2^2 - 1)] = 0,$$

and $\eta > 0$ is a known constant that measures the nonlinearity. We are interested in testing the hypotheses

$$H_0 : c'\theta = \theta_1 = 1 \text{ v.s. } H_1 : c'\theta = \theta_1 \neq 1.$$

Below I use $Z = (Z_1, Z_2, Z_3) \sim N(0, I_3)$. Under H_0 with $\mu = 0$, the identified set is $\Theta_I(1) = \{(1, 1), (1, -1)\}$. Under local alternatives,

$$J_n(r) = \min_{\theta_2 \in [-2, 2]} n \left[\frac{\bar{W}_1 - 1 + \theta_2}{\hat{\sigma}_1} \right]_-^2 + n \left[\frac{\bar{W}_2 - 1 - \theta_2}{\hat{\sigma}_2} \right]_-^2 + n \left(\frac{\bar{W}_3 + \eta(\theta_2^2 - 1)}{\hat{\sigma}_3} \right)^2 \\ \xrightarrow{d} \min_{j=1,2} \frac{[2\eta(Z_j - \mu) + Z_3]_-^2}{4\eta^2 + 1}$$

where $\frac{[2\eta(Z_1 - \mu) + Z_3]_-^2}{4\eta^2 + 1}$ is the local minimum around $\theta = (1, -1)$ and $\frac{[2\eta(Z_2 - \mu) + Z_3]_-^2}{4\eta^2 + 1}$ is the local minimum around $\theta = (1, 1)$. If the two local minimums are both zero, we have two minimizers, otherwise, the minimizer is unique. The calculation details are given in Appendix C.

Test PR. We can show that

$$J_n^{\text{PR}}(1) | W_n \xrightarrow{d} \min_j \frac{[2\eta Z_j + Z_3]_-^2}{4\eta^2 + 1}.$$

Test LL. In this simple example, I use

$$\hat{\Theta}_I(r) = \arg \min_{\theta \in \Theta(r)} n \left[\frac{\bar{W}_1 - 1 + \theta_2}{\hat{\sigma}_1} \right]_-^2 + n \left[\frac{\bar{W}_2 - 1 - \theta_2}{\hat{\sigma}_2} \right]_-^2 + n \left(\frac{\bar{W}_3 + \eta(\theta_2^2 - 1)}{\hat{\sigma}_3} \right)^2$$

which is the set of all minimizers.³ The bootstrapped test statistic $J_n^{\text{LL}}(1)$ depends on $\hat{\Theta}_I(1)$. If $\hat{\Theta}_I(1)$ is singleton with $\hat{\theta}_2$ close to -1 , we let $\delta_n = 1$ and

$$J_n^{\text{LL}}(1) | \{\delta_n = 1, W_n\} \xrightarrow{d} \frac{[2\eta Z_1 + Z_3]_-^2}{4\eta^2 + 1}.$$

If $\hat{\Theta}_I(1)$ is singleton with $\hat{\theta}_2$ close to 1 , we let $\delta_n = 2$ and

$$J_n^{\text{LL}}(1) | \{\delta_n = 2, W_n\} \xrightarrow{d} \frac{[2\eta Z_2 + Z_3]_-^2}{4\eta^2 + 1}.$$

And if $\hat{\Theta}_I(1)$ contains more than one elements, we let $\delta_n = 3$ and

$$J_n^{\text{LL}}(1) | \{\delta_n = 3, W_n\} \xrightarrow{d} \min_{j=1,2} \frac{[2\eta Z_j + Z_3]_-^2}{4\eta^2 + 1}.$$

See Appendix C for a specific form of δ_n .

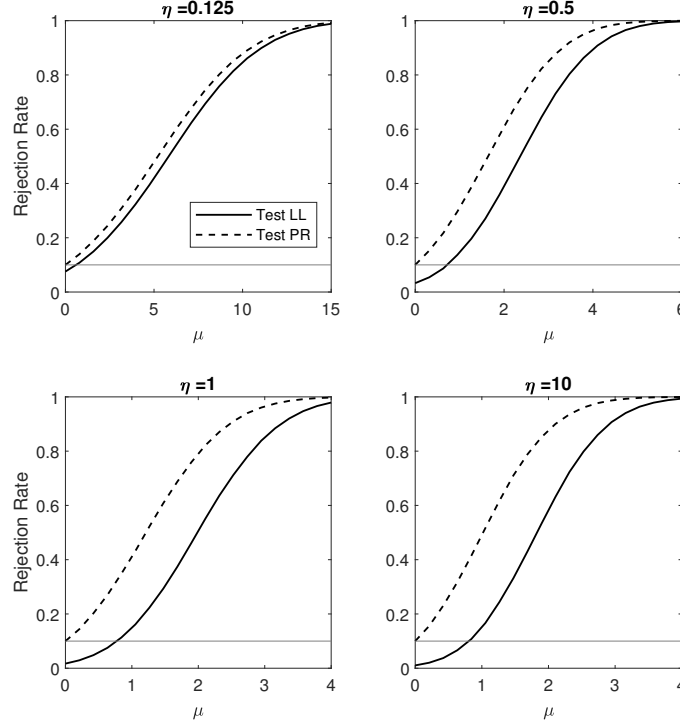
For all $\eta > 0$, the asymptotic distribution of $J_n^{\text{LL}}(1)$ first order stochastically dominates $J_n^{\text{PR}}(1)$ and Test PR has strictly higher power than Test LL. The power loss comes from the fact that we are unable to consistently estimate $\Theta_I(r)$ at a fast enough rate, and thus, with positive probability, $\hat{\Theta}_I(r)$ can only approximate a strict subset of $\Theta_I(r)$. In Figure 1, I plot the local power of Test LL and Test PR for η from 0.125 to 10. We can see that the power loss increases with the nonlinearity η , especially for η smaller than one. However, when η changes from one to ten, the power curves do not change significantly, and thus the power loss stabilizes for large η .

3.4 Computational properties

To test $H_0 : c'\theta_0 = r$ for a given r , the most computationally demanding step in BCS17 is Test PR. First, Test PR bootstraps a statistic that minimizes a nonlinear objective function. Second, we need to calculate $v_{n,j}^*(\theta)$ for (a large number of) $\theta \in \Theta(r)$. This is computationally expensive as it requires evaluating the moment function $m(W_i, \theta)$ for all observations $i = 1, \dots, n$. Test LL addresses the first point by approximating the objective function and the constraints by linear counterparts so that the inner minimization problem in (6) is simply quadratic programming.

³The asymptotic power is the same if we use $\hat{\Theta}_I(r)$ which includes only one minimizer. The reason is that if there are two distinct minimizers, then $J_n(r) = 0$ with probability approaching one. In this case, we do not reject H_0 and the critical value does not matter.

Figure 1: Example 2 local power with coverage rate $1 - \alpha = 0.9$



And Test LL addresses the second point by conducting the outer minimization in (5) over $\hat{\Theta}_I(r)$, so that we only need to calculate $v_{n,j}^*(\theta)$ for $\theta \in \hat{\Theta}_I(r)$, which is much smaller than $\Theta(r)$.

Note that both Test DR and Test LL involve a potentially nonlinear constraint $\hat{\Theta}_I(r)$, which seems computationally costly. However, $\hat{\Theta}_I(r)$ is not required to contain all the minimizers and the number of elements in $\hat{\Theta}_I(r)$ trades off the computation and power properties. In practice, $\hat{\Theta}_I(r)$ usually contains a finite number of points, and taking minimization over $\hat{\Theta}_I(r)$ can be computationally easy.

For both Test LL and BCS, the confidence intervals are constructed by inverting the test over a grid of r and the computing time increases linearly in the number of grid points. BCS17 calculate the test on evenly spaced points in their simulation. An alternative method is the E-A-M algorithm proposed in KMS19, which improves the implementation of BCS17 by a factor of roughly 4 in the KMS19 simulation. In Section 5, I implement both Test BCS and Test LL with the E-A-M algorithm.

The computational gain of Test LL comes at the cost of potential power loss and additional user-chosen tuning parameters. To compute the critical value, Test PR searches over the null set $\Theta(r)$ while Test LL only searches locally around $\hat{\Theta}_I(r)$. Thus we may sacrifice the power compared to Test PR if $\hat{\Theta}_I(r)$ includes only a few elements or if it can not be a consistent estimator for $\Theta_I(r)$, especially when the dimension of θ is relatively high and $\Theta_I(r)$ is large. As for the tuning parameters, in addition to the GMS parameter κ_n , which is also used in BCS, LL needs (i) the penalty parameter λ_n and (ii) the set $\Delta_n^r(\theta)$ which is a local approximation of $\sqrt{n}(\Theta(r) - T^r\theta)$. To reduce the number of tuning parameters, I suggest setting $\lambda_n = \sqrt{n}\kappa_n^2$.

and constructing $\Delta_n^r(\theta)$ based on shrinking $\Theta(r)$ by $a\varepsilon_n \propto \sqrt{\frac{\ln \kappa_n^2}{n}}$, see Section 4.1 for detailed discussion.

4 Implementation

In this section, I illustrate the implementation details of constructing $\Delta_n^r(\theta)$ in (6), $\hat{\Theta}_I(r)$ in (5), and give a step by step algorithm to construct the confidence interval.

4.1 Examples of $\Delta_n^r(\theta)$

Recall that Assumption 4 requires that for some $a > 0$,

$$0 \in \Delta_n^r(\theta) \subseteq \sqrt{n}(\Theta(r) - T^r\theta)$$

for all $\theta \in \Theta(r) \cap \Theta_I(r, F)^{a\varepsilon_n}$ with $\varepsilon_n = n^{-1/2}(\ln \kappa_n^2)^{\frac{1}{\chi}}$ with probability approaching one. Note that

$$\sqrt{n}(\Theta(r) - T^r\theta) = \sqrt{n}(\Theta(r) - \theta) + \sqrt{n}(\theta - T^r\theta),$$

thus the main idea is to shrink the boundary of $\sqrt{n}(\Theta(r) - \theta)$ by an “upper bound” of $\sqrt{n}(\theta - T^r\theta)$.

First, for a general Θ , define

$$\overline{\Theta(r)} = \left\{ \theta \in \mathbb{R}^{d_\theta} : c'\theta = r, \theta \notin \Theta \right\}$$

to be the set of parameters satisfying the subvector restriction but not in Θ , and define

$$\Theta(r)^{-\varepsilon_n} = \left\{ \theta \in \Theta(r) : d\left(\theta, \overline{\Theta(r)}\right) \geq \varepsilon_n \right\}$$

to be the set of parameters in $\Theta(r)$ but away from the boundary of Θ . A valid choice is

$$\Delta_n^r(\theta) = \sqrt{n}(\Theta(r)^{-\varepsilon_n} - \theta) \cup \{0_{d_\theta}\}. \quad (24)$$

Assumption 4.1 holds trivially. To verify Assumption 4.2. For all $\theta \in \Theta(r) \cap \Theta_I(r, F)^{\varepsilon_n}$, $\|T^r\theta - \theta\| \leq \varepsilon_n$, thus $\Theta(r)^{-\varepsilon_n} \subseteq \Theta(r) - (T^r\theta - \theta)$, which implies that

$$\sqrt{n}(\Theta(r)^{-\varepsilon_n} - \theta) \subseteq \sqrt{n}(\Theta(r) - (T^r\theta - \theta) - \theta) = \sqrt{n}(\Theta(r) - T^r\theta). \quad (25)$$

Example 3. Let $\Theta = \{\theta : \underline{\theta}_j \leq \theta_j \leq \bar{\theta}_j, j = 1, \dots, v\}$. We are interested in testing $H_0 : c'\theta = r$ with $c = (1, 0_{v-1})'$. In this sample,

$$\begin{aligned} \Theta(r) &= \{(r, \theta_2, \dots, \theta_v) : \underline{\theta}_j \leq \theta_j \leq \bar{\theta}_j, \quad j = 2, \dots, v\}, \\ \overline{\Theta(r)} &= \{(r, \theta_2, \dots, \theta_v) : \theta_j \in (-\infty, \underline{\theta}_j) \cup (\bar{\theta}_j, +\infty), \quad j = 2, \dots, v\}, \\ \Theta(r)^{-\varepsilon_n} &= \{r\} \times \prod_{j=2}^v [\underline{\theta}_j + \varepsilon_n, \bar{\theta}_j - \varepsilon_n]. \end{aligned}$$

And (24) is equivalent to

$$\Delta_n^r(\theta) = \left(\{0\} \times \prod_{j=2}^v [\sqrt{n}(\underline{\theta}_j - \theta_j + \varepsilon_n), \sqrt{n}(\bar{\theta}_j - \theta_j - \varepsilon_n)] \right) \cup \{0_v\}. \quad (26)$$

We can also scale ε_n in (26) so that the test is invariant to the unit of parameters. That is

$$\Delta_n^r(\theta) = \left(\{0\} \times \prod_{j=2}^v [\sqrt{n}(\underline{\theta}_j - \theta_j + \epsilon_{n,j}), \sqrt{n}(\bar{\theta}_j - \theta_j - \epsilon_{n,j})] \right) \cup \{0_v\} \quad (27)$$

with $\epsilon_{n,j} = \frac{\bar{\theta}_j - \underline{\theta}_j}{2} \varepsilon_n$. In (27), Assumption 4 holds with $a = \min_{j=2,\dots,v} \frac{\bar{\theta}_j - \underline{\theta}_j}{2}$.

If Θ is a polytope, i.e. $\Theta = \{\theta : A_j \theta \leq b_j, j = 1, \dots, J\}$ where A_j is a $1 \times d_\theta$ row vector, then we can use

$$\Delta_n^r(\theta) = \{\Delta : c' \Delta = 0, A_j \Delta \leq \sqrt{n} \max(b_j - A_j \theta - \|A_j\| \varepsilon_n, 0), j = 1, \dots, J\}. \quad (28)$$

It is straightforward to verify that $0 \in \Delta_n^r(\theta)$ for all $\theta \in \Theta(r)$ and $\Delta_n^r(\theta) = \sqrt{n}(\Theta(r) - T^r \theta)$ for all $\theta \in \Theta(r) \cap \Theta_I(r, F)^{\varepsilon_n}$. I illustrate (28) with the following two examples.

Example 4. Let $\Theta = \{(\theta_1, \theta_2, \theta_3) : \theta_1 \geq 0, \theta_2 \geq 0, \theta_1 + \theta_2 \leq 1, 0 \leq \theta_3 \leq 1\}$. We are interested in testing $H_0 : c' \theta = r$ with $c = (0, 0, 1)$ and $r = 0$. Assume $\Theta_I(r) = \{(\frac{1}{2}, \theta_2, 0) : 0 \leq \theta_2 \leq \frac{1}{2}\}$. In this case, $A_1 = (-1, 0, 0)$, $A_2 = (0, -1, 0)$, $A_3 = (1, 1, 0)$, $A_4 = (0, 0, 1)$, $A_5 = (0, 0, -1)$ and $b = (0, 0, 1, 1, 0)$. $\Theta(r) = \{(\theta_1, \theta_2, 0) : \theta_1 \geq 0, \theta_2 \geq 0, \theta_1 + \theta_2 \leq 1\}$.

For $(\theta_1, \theta_2, 0) \in \Theta_I(r, F)^{\varepsilon_n}$, we can easily calculate $\Delta_n^r(\theta)$ defined in (28)

$$\Delta_n^r(\theta) = \{(\Delta_1, \Delta_2, 0) : \Delta_1 \geq \sqrt{n} \min(\varepsilon_n - \theta_1, 0), \Delta_2 \geq \sqrt{n} \min(\varepsilon_n - \theta_2, 0), \\ \Delta_1 + \Delta_2 \leq \sqrt{n} \max(1 - \sqrt{2} \varepsilon_n - \theta_1 - \theta_2, 0)\}.$$

If (θ_1, θ_2) is far away from the boundary of parameter space, for instance, $(\theta_1, \theta_2) = (\frac{1}{2} - \varepsilon_n, \frac{1}{4})$ in Figure 2a, then with n large enough,

$$\Delta_n^r(\theta) = \{(\Delta_1, \Delta_2, 0) : \Delta_1 \geq \sqrt{n}(\varepsilon_n - \theta_1), \Delta_2 \geq \sqrt{n}(\varepsilon_n - \theta_2), \\ \Delta_1 + \Delta_2 \leq \sqrt{n} (1 - (\sqrt{2} \varepsilon_n + \theta_1 + \theta_2))\}.$$

This is $\sqrt{n}(\Theta(r) - \theta)$ with each side of the boundary shrunk by width $\sqrt{n} \varepsilon_n$, see Figure 2c. However, if (θ_1, θ_2) is close to the boundary of parameter space, for instance, $(\theta_1, \theta_2) = (\frac{1}{2} - \frac{\varepsilon_n}{2}, \frac{1}{2})$, then with n large enough,

$$\Delta_n^r(\theta) = \{(\Delta_1, \Delta_2, 0) : \Delta_1 \geq \sqrt{n}(\varepsilon_n - \theta_1), \Delta_2 \geq \sqrt{n}(\varepsilon_n - \theta_2), \Delta_1 + \Delta_2 \leq 0\}.$$

In this case, the hypotenuse boundary is shrunk by $\frac{\sqrt{n} \varepsilon_n}{2\sqrt{2}}$. The intuition is that θ is close to the hypotenuse boundary, thus $T^r \theta$ can not be too far away along direction $(1, 1)$. Thus the shrinkage of this bound is smaller, see Figure 2f.

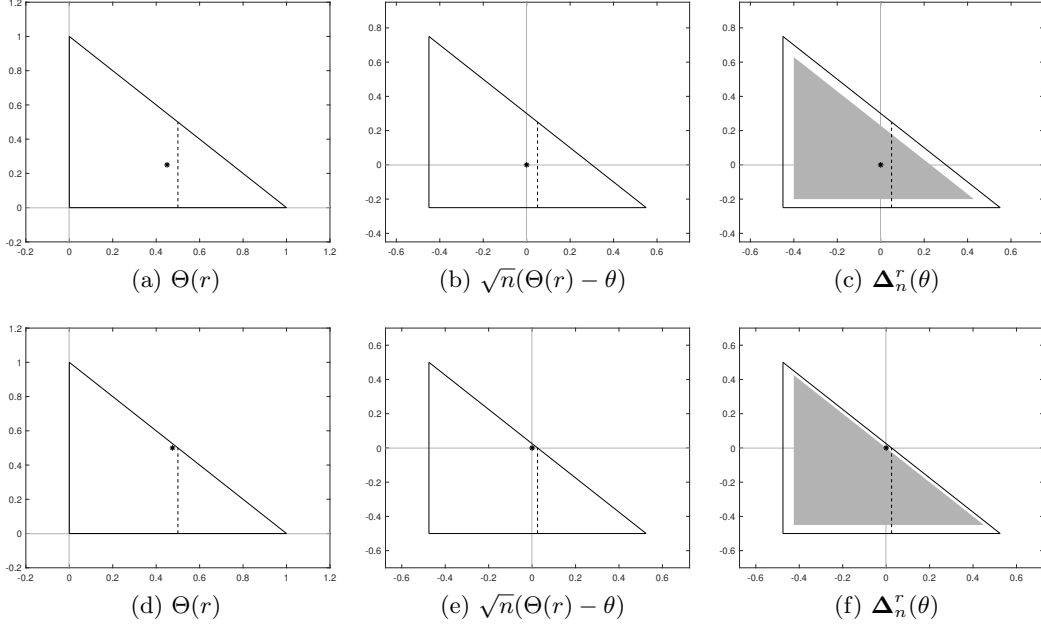


Figure 2: Illustration of $\Delta_n^r(\theta)$ in Example 4

2a is the cross-section plot for the (θ_1, θ_2) at $\theta_3 = 0$. The horizontal axis is θ_1 , the vertical axis is θ_2 and the unit is one. The “*” dot is $\theta = (\frac{1}{2} - \varepsilon_n, \frac{1}{4})$. The solid triangle is $\Theta(r)$ and the vertical dash line is $\Theta_I(r)$. **2b** is **2a** recentered at θ and scaled by $\sqrt{n}$. The unit for two axes is $\sqrt{n}$. In **2c** the shaded triangle is $\Delta_n^r(\theta)$ and the rest is the same as **2b**. **2d**, **2e** and **2f** are the same, except $(\theta_1, \theta_2) = (\frac{1}{2} - \frac{\varepsilon_n}{2}, \frac{1}{2})$.

Example 3 (continued). In the spirit of (27) and (28), it is also valid to use

$$\Delta_n^r(\theta) = \{0\} \times \prod_{j=2}^v [\min(\sqrt{n}(\underline{\theta}_j - \theta_j + \epsilon_{n,j}), 0), \max(\sqrt{n}(\bar{\theta}_j - \theta_j - \epsilon_{n,j}), 0)] \quad (29)$$

with $\epsilon_{n,j} = \frac{\bar{\theta}_j - \underline{\theta}_j}{2} \varepsilon_n$.

4.2 Construction of $\hat{\Theta}_I(r)$

As discussed in Remark 1, we can construct $\hat{\Theta}_I(r)$ by

$$\hat{\Theta}_I(r) \subseteq \left\{ \theta \in \Theta(r) : S(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta)) \leq J_n(r) + \delta_n \right\} \quad (30)$$

with $\delta_n \rightarrow 0$ being the tolerance for “approximate minimum”. The full set on the right hand side of (30) is defined by non-linear restrictions and may contain uncountably many points, but the suggested choice of $\hat{\Theta}_I(r)$ is a subset with only a finite number of “approximate minimizers”. In practice, $\hat{\Theta}_I(r)$ can be the set of minimizers in (5) generated by different initial values. For instance, we can run the MultiStart minimization algorithm in MATLAB with $\bar{K} \in \mathbb{N}$ different initial values, where $\bar{K}$ is the maximum number of elements. The algorithm will return the minimum $J_n(r)$ and $K \leq \bar{K}$ distinct minimizers. This method avoids grid search or nonlinear restrictions.

Note that when $J_n^{\text{LL}}(r)$ is large, we can further improve computational efficiency by starting with a singleton $\hat{\Theta}_I(r)$. Specifically, we first calculate $J_n(r)$ and simultaneously get at least

one minimizer $\hat{\theta}_{n,r}$. Then we can use this single value to calculate the critical value cv_1^{LL} . If $J_n(r) > cv_1^{LL}$, there is no need to search for more elements in $\hat{\Theta}_I(r)$ and recalculate the critical value, since the null hypothesis is rejected anyway. If $J_n(r) \leq cv_1^{LL}$, we use MultiStart with $\bar{K}$ initial values, and use $\hat{\Theta}_I(r)$ as the collection of K minimizers and recalculate cv_K^{LL} . The test result is given by

$$\phi(r) = \mathbf{1} [J_n^{LL}(r) > cv_K^{LL}].$$

In Section 4.3, I give a detailed algorithm on how to construct $\hat{\Theta}_I(r)$.

Remark 3. It plays an important role in the scalability of Test LL that the number of elements in $\hat{\Theta}_I(r)$ is finite and does not grow rapidly with d_θ . $\hat{\Theta}_I(r)$ is chosen by the researcher depending on the dimension of parameters, computational ability, and desired power. This allows the researcher to balance flexibly the computational time and the length of the confidence interval, which is not an obvious option for the existing methods. When the dimension of parameters is relatively large, we may use a larger $\bar{K}$ to improve power. Moreover, the computing time increases much slower than a multiple of $\bar{K}$ for the following two reasons. First, as discussed in the previous paragraph, a singleton $\hat{\Theta}_I(r)$ is sufficient when $J_n(r) > cv_1^{LL}(r)$. Thus $\hat{\Theta}_I(r)$ with $K > 1$ elements is only needed for r close to or inside the confidence interval. Second, the MultiStart minimization algorithm and the calculation of $\Delta^{LL}(\theta)$ in (23) can be easily parallelized. Therefore, the LL inference procedure has scalability with the sample and model size even with a relatively large $\hat{\Theta}_I(r)$.

4.3 A step by step algorithm

In this section, I give step by step algorithm to compute the confidence interval by inverting Test LL. For completeness, I list below the notation used in this section. Define

$$\begin{aligned} \bar{m}_n(\theta) &:= \frac{1}{n} \sum_{i=1}^n m(W_i, \theta), \\ \hat{\Sigma}_n(\theta) &:= \frac{1}{n} \sum_{i=1}^n (m(W_i, \theta) - \bar{m}_n(\theta)) (m(W_i, \theta) - \bar{m}_n(\theta))', \\ \hat{D}_n(\theta) &:= \text{diag}(\hat{\Sigma}_n(\theta)), \\ \hat{\Omega}_n(\theta) &:= \hat{D}_n(\theta)^{-\frac{1}{2}} \hat{\Sigma}_n(\theta) \hat{D}_n(\theta)^{-\frac{1}{2}}. \end{aligned} \tag{31}$$

Let $\kappa_n = \sqrt{\ln(n)}$, $\lambda_n = \sqrt{n} \kappa_n^2$, $\varepsilon_n = \sqrt{\frac{\ln \kappa_n^2}{n}}$ and set δ_n to be a small non-negative number, e.g. $\min\{\frac{1}{n}, 10^{-4}\}$, for the tolerance of the approximate minimizer set $\hat{\Theta}_I(r_h)$ in (30) and used in Step 4. In addition, let

$$S(m, \Sigma) = \sum_{j=1}^p \left[m_j / \Sigma_{jj}^{\frac{1}{2}} \right]_-^2 + \sum_{j=p+1}^k \left[m_j / \Sigma_{jj}^{\frac{1}{2}} \right]^2.$$

Define $\Delta_n^r(\theta)$ satisfying Assumption 4, for example,

$$\Delta_n^r(\theta) = \sqrt{n}(\Theta(r)^{-\varepsilon_n} - \theta) \cup \{0_{d_\theta}\}.$$

I assume there is a grid search algorithm R that takes $\{r_j, J_n(r_j), cv_n^{LL}(r_j, 1 - \alpha)\}_{j=1}^h$ as input and returns r_{h+1} and s_{h+1} as the next evaluating point and a stopping indicator, respectively.

If $s_{h+1} = 1$ then stop the grid search. For instance, the simplest evenly spaced grid search with $K + 1$ grid points over $[\underline{r}, \bar{r}]$ is given by

$$R\left(\{r_j, J_n(r_j), cv_n^{LL}(r_j, 1 - \alpha)\}_{j=1}^h\right) := \left[\underline{r} + \frac{\bar{r} - \underline{r}}{K} \times (h - 1), \mathbf{1}[h \geq K + 1]\right].$$

The E-A-M algorithm in KMS19 approximates the difference between the test statistic and critical value through a probabilistic model that is updated during the grid search, and selects the most relevant points as the next point to evaluate. The algorithm stops when the grid search converges, or the maximum iteration is achieved.

1. Initialize $h = 1$, $r_1 \in c'\Theta$ and $s_1 = 0$.
2. Estimate the test statistic by minimizing the test function

$$J_n(r_h) = \inf_{\theta \in \Theta(r_h)} S(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta)),$$

and get minimizer $\hat{\theta}_{n,r_h}$. Let $\hat{\Theta}_I(r_h) = \{\hat{\theta}_{n,r_h}\}$ and $Singleton = 1$.⁴

3. Calculate the critical value for r_n

(a) simulate ζ_i^b i.i.d for $i = 1, \dots, n$, $b = 1, \dots, B$, such that

$$\zeta_i^b | \{W_i\}_{i=1}^n \sim N(0, 1).$$

(b) for each b , calculate

$$J_n^{LL,b}(r) = \inf_{\theta \in \hat{\Theta}_n(r_h)} S\left(v_n^{*,b}(\theta) + \varphi(\xi_n(\theta)) + \hat{G}_n(\theta) \tilde{\Delta}(\theta), \hat{\Omega}_n(\theta)\right),$$

$$\text{where } v_n^{*,b}(\theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{D}_n^{-\frac{1}{2}}(\theta) (m(W_i, \theta) - \bar{m}_n(\theta)) \zeta_i^b,$$

$$\varphi_j(\xi_n(\theta)) = \begin{cases} \left[\frac{\sqrt{n} \bar{m}_{n,j}(\theta)}{\kappa_n \hat{\sigma}_{n,j}(\theta)} \right]_+ & \text{if } j = 1, \dots, p \\ 0 & \text{if } j = p + 1, \dots, k, \end{cases}$$

and $\hat{G}_n$ is a consistent estimator for $\nabla_{\theta'} \left(D_F(\theta)^{-\frac{1}{2}} E_F(m(W_i, \theta)) \right)$, and

$$\Delta(\theta) \in \operatorname{argmin}_{\Delta \in \mathbf{\Delta}_n^r(\theta)} \left(S\left(v_n^{*,b}(\theta) + \varphi(\xi_n(\theta)) + \hat{G}_n(\theta) \Delta, \hat{\Omega}_n(\theta)\right) + \frac{\lambda_n}{n} \|\Delta\|^2 \right).$$

(c) get the $1 - \alpha$ quantile of $\left\{ J_n^{LL,b}(r_h) \right\}_{b=1}^B$, and denoted it as $cv_n^{LL}(r_h, 1 - \alpha)$.

4. If $Singleton = 1$ and $J_n(r_h) \leq cv_n^{LL}(r_h, 1 - \alpha)$,

⁴The procedure to start with a singleton set is a pure computational device and does not suffer from a pre-test distortion.

- (a) run a minimization algorithm with $\bar{K}$ initial values (e.g. MultiStart in MATLAB). Let $\tilde{\Theta}_I(r_h)$ be the set of $\bar{K}$ returned minimizers and

$$\hat{\Theta}_I(r_h) = \left\{ \theta \in \tilde{\Theta}_I(r_h) : S(\hat{D}_n(\theta)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta), \hat{\Omega}_n(\theta)) \leq J_n(r_h) + \delta_n \right\}.$$

$\hat{\Theta}_I(r_h)$ is the set of approximate minimizers and it rules out the local minimizers far from the global minimizer.

- (b) let $Singleton = 0$
(c) go to Step 3

Else,

- (a) go to Step 5

5. Let

$$[r_{h+1}, s_{h+1}] = R \left(\{r_j, J_n(r_j), cv_n^{LL}(r_j, 1 - \alpha)\}_{j=1}^h \right).$$

6. If $s_{h+1} = 0$, let $h = h + 1$ and return to Step 2. If $s_{h+1} = 1$, return the confidence interval

$$\left[\min_{j \leq h: J_n(r_j) \leq cv_n^{LL}(r_j, 1 - \alpha)} r_j, \max_{j \leq h: J_n(r_j) \leq cv_n^{LL}(r_j, 1 - \alpha)} r_j \right].$$

5 Monte Carlo Simulations

In this section, I study the size and power properties by Monte Carlo simulations of a simple two-player entry game.

5.1 Model

There are two firms $j = 1, 2$ choosing whether to enter a market $i \in \{1, \dots, n\}$. The payoff is given by

$$\pi_{j,i} = A_{j,i}(e_{j,i} + \delta_j A_{-j,i} + \sum_{q=1}^{d_X} \beta_{j,q} X_{q,i}),$$

where $A_{j,i} = 1$ if j chooses to enter market i . There are d_X type markets and $X_{q,i}$ are observed binary market type indicators with distribution $P(X_{q,i} = 1) = p_q = \frac{1}{d_X}$. Assume p_q is known for simplicity. Let

$$(e_{1,i}, e_{2,i}) | X_i \sim N \left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

The moment conditions are

$$\begin{aligned}
E_F[m_{1,q}(W_i, \theta)] &= E_F[G_\rho([-\beta_{1,q}, \infty) \times (-\infty, -\beta_{2,q} - \delta_2]) p_q - A_{1,i}(1 - A_{2,i})X_{q,1}] \geq 0 \\
E_F[m_{2,q}(W_i, \theta)] &= E_F[A_{1,i}(1 - A_{2,i})X_{q,1} - G_\rho([-\beta_{1,q}, \infty) \times (-\infty, -\beta_{2,q} - \delta_2]) p_q \\
&\quad + G_\rho([-\beta_{1,q}, -\beta_{1,q} - \delta_1) \times (-\beta_{2,q}, -\beta_{2,q} - \delta_2]) p_q] \geq 0 \\
E_F[m_{3,q}(W_i, \theta)] &= E_F[(1 - A_{1,i})(1 - A_{2,i})X_{q,1} - G_\rho((-\infty, -\beta_{1,q}] \times (-\infty, -\beta_{2,q}]) p_q] = 0 \\
E_F[m_{4,q}(W_i, \theta)] &= E_F[A_{1,i}A_{2,i}X_{q,1} - G_\rho([-\beta_{1,q} - \delta_1, \infty) \times [-\beta_{2,q} - \delta_2, \infty)) p_q] = 0
\end{aligned}$$

where $W_i = (A_i, X_i)$ and $G_\rho(A) = P((e_1, e_2) \in A)$.

To examine the performance of the inference procedures under different model sizes, I conduct the simulation with $d_X = 2, 4, 6$, where $d_\theta = 7, 11, 15$, the number of moment inequalities $p = 4, 8, 12$ and the number of total moment conditions $k = 8, 16, 24$, respectively. The true parameter is $\theta = (\beta_{1,1}, \dots, \beta_{1,d_X}, \beta_{2,1}, \dots, \beta_{2,d_X}, \delta_1, \delta_2, \rho)$ where

$$\begin{aligned}
\beta_1 &= (0.1, \quad 1.1, \quad -0.1, \quad -0.5, \quad 0.8, \quad -0.5), \\
\beta_2 &= (0.3, \quad -0.8, \quad 0, \quad -0.5, \quad -0.9, \quad -0.2), \\
\delta_1 &= -0.1, \quad \delta_2 = -0.3, \quad \rho = 0.4.
\end{aligned}$$

In the region of multiple equilibria, the probability of selecting $A_i = (1, 0)$ is 0.5. The parameter space is

$$\begin{aligned}
\Theta &= \left\{ \theta = (\beta_1, \beta_2, \delta_1, \delta_2, \rho) \in \mathbb{R}^{2d_X+3} : \beta_{j,q} \in [-2, 2], \delta_j \in [-2, 0] \right. \\
&\quad \left. \text{for all } j = 1, 2, q \in \{1, \dots, d_X\}, \rho \in [0, 0.85] \right\}.
\end{aligned}$$

We are interested in constructing a 95% confidence interval (CI) for θ_1 . The identified set of θ_1 is reported in the first row of Table 1a. Additional simulation results for ρ are included in Appendix D.

5.2 Implementation details

I run all methods with $MC = 1,000$ replications and $B = 600$ bootstrap draws on a server with 16 workers.⁵ For Test BCS, I use $\kappa_n = \sqrt{\log(n)}$ as recommended by Andrews and Soares (2010), the MMM test function and $\varphi^{(1)}$ in (18) as suggested in BCS17. As for Test LL, I use $\lambda_n = \sqrt{n}\kappa_n^2$, and

$$\Delta_n^r(\theta) = \{0\} \times \prod_{j=2}^{d_\theta} [\min(0, \sqrt{n}(\underline{\theta}_j - \theta_j + \epsilon_{n,j}), \max(0, \sqrt{n}(\bar{\theta}_j - \theta_j + \epsilon_{n,j}))]. \quad (32)$$

with $\epsilon_{n,j} = \frac{(\bar{\theta}_j - \underline{\theta}_j)}{2}\epsilon_n$ and $\epsilon_n = \sqrt{\frac{\ln(\kappa_n^2)}{n}}$ as in (29). I use the E-A-M algorithm to calculate the confidence intervals for both profiling methods. As for $\hat{\Theta}_I(r)$, I follow BCS17 and calculate the test statistic by running Matlab command *fmincon* twice with two distinct initial values, and each run returns a minimizer. $\hat{\Theta}_I(r)$ is the set of distinct minimizers, which has at most two

⁵BCS17 set $B = 301$ and KMS19 report the results with both $B = 301$ and $B = 1001$. $B = 600$ is a reasonable choice which makes the Monte Carlo simulations computationally feasible. In practice, researchers can use a larger B , e.g. $B = 1000$, to get a more accurate critical value.

elements. Test LL and Test BCS use the same $\hat{\Theta}_I(r)$. In general, larger $\bar{K}$ gives (weakly) larger power at the cost of increasing computing time. The recommended range of $\bar{K}$ is between two to ten, depending on the model size and computational power.

The calibrated projection-based test proposed in KMS has form

$$\left[\inf_{J_n^{\text{KMS}}(\theta) \leq cv_n^{\text{KMS}}(\theta)} c'\theta, \sup_{J_n^{\text{KMS}}(\theta) \leq cv_n^{\text{KMS}}(\theta)} c'\theta \right],$$

where $J_n^{\text{KMS}}(\theta)$ and $cv_n^{\text{KMS}}(\theta)$ are the KMS test statistic and critical value evaluated at $\theta \in \Theta$. Let m be the number of initial points to use in multi-start, and I calculate the KMS confidence interval using the E-A-M algorithm with three different m . The recommended value of m is between 10 and 100. I run the algorithm with $m = 20, 40, 80$ for each sample and get three confidence intervals. Note that the seeds to generate the original and bootstrapped data are fixed, thus the difference among these three confidence intervals only comes from the randomness in the E-A-M algorithm. It turns out that the E-A-M algorithm can be sensitive to the choice of m , especially when d_θ is large. For instance, for $n = 1,000$, $d_X = 6$, the difference between the three lower bounds is larger than 0.1 for around 16% of the samples (with the median length of CI around 1). It is also unclear which is the optimal m . For 19% (resp. 19% , 14%) of the samples, $m = 20$ (resp. $m = 40$, $m = 80$) gives a strictly smaller lower bound than the other two by at least 0.005. Therefore, implementing the E-A-M algorithm with different values of m can reduce sensitivity.⁶ I take the union of these three confidence intervals as the final one.

5.3 Simulation results

In this section, I present the simulation results.

Confidence intervals and coverage rates

The median 95% CI and coverage rate are reported in Table 1. The median CI is the median of the lower bounds to the median of the upper bounds. The ‘‘Upper’’ coverage rate is for the upper bound of the identified set of θ_1 , and similarly for ‘‘Lower’’ coverage. The median confidence intervals of LL and BCS are similar, and the coverage rates are higher than the nominal value for all model and sample sizes. The coverage rates for KMS with $N = 4,000$, $d_X = 4, 6$ and $N = 1,000$, $d_X = 4$ are close to, but slightly lower than, the nominal rate. It could be a result of the sensitivity of the E-A-M algorithm. The KMS median confidence intervals are strictly shorter than BCS and LL.

Computing time

The average computing time is reported in Table 2. The computing time of BCS is in the second and third rows, where ‘‘add’’ represents for ‘‘additively separable model’’ and ‘‘gen’’ represents

⁶It may also help to try a combination of different values of other tuning parameters, e.g. the number of initial points and points to keep from expected improvement maximization. Here I keep other tuning parameters at the default values because of the limit of computation.

Table 1: Simulation Results: Median CI and Coverage

		(a) Median 95% CI					
		$d_X = 2$		$d_X = 4$		$d_X = 6$	
Identified Set		[0.044, 0.265]		[0.064, 0.167]		[0.071, 0.159]	
$N = 1,000$	LL	[-0.134, 0.817]		[-0.294, 0.926]		[-0.489, 1.052]	
	BCS	[-0.149, 0.842]		[-0.282, 0.922]		[-0.404, 0.997]	
	KMS	[-0.078, 0.737]		[-0.158, 0.795]		[-0.250, 0.839]	
$N = 4,000$	LL	[-0.037, 0.721]		[-0.104, 0.773]		[-0.177, 0.823]	
	BCS	[-0.046, 0.737]		[-0.110, 0.780]		[-0.164, 0.816]	
	KMS	[-0.013, 0.690]		[-0.048, 0.720]		[-0.083, 0.742]	

		(b) Coverage Rate					
		$d_X = 2$		$d_X = 4$		$d_X = 6$	
		Lower	Upper	Lower	Upper	Lower	Upper
$N = 1,000$	LL	99.3	99.9	99.7	100.0	100.0	100.0
	BCS	99.4	99.9	99.3	100.0	98.9	100.0
	KMS	96.6	97.7	94.4	97.7	96.8	98.3
$N = 4,000$	LL	98.4	99.7	99.6	100.0	99.5	100.0
	BCS	99.2	99.6	99.3	99.9	99.3	100.0
	KMS	96.5	97.5	91.6	98.7	92.3	98.7

for “general model”. Note that in this simulation, the data and the parameters are additively separable in the moment conditions, and so $v_n^*(\theta)$ is invariant to θ . “BCS-add” exploits this feature and calculates $v_{n,j}^*(\theta)$ only once for each bootstrap. I also report the computing time “BCS-gen” using BCS17 code that works with general models, which calculates $v_n^*(\theta)$ at each θ . “BCS-gen” is more relevant in a non-separable model with similar sample and model sizes. Note that the confidence intervals calculated by these two methods are numerically the same. The computing time of KMS is reported in the fourth row. For each sample, I run the KMS method with three choices of m , and the reported time is the smallest one corresponding to the largest confidence interval.⁷

LL is computationally attractive and takes less than 81 seconds in all models. BCS-gen is very time-consuming and especially sensitive to the sample size. LL improves upon BCS-gen by a factor of up to 170 ($d_X = 6$, $n = 4,000$), where a factor of 9 comes from using a local search to reduce the time evaluating $v_n^*(\theta)$ (compare BCS-gen and BCS-add), and a further factor of 19 from the linearization of moment function (compare BCS-add and LL). The computing time of BCS-add and KMS remains approximately the same as n increases but increases significantly with the model size d_X .

LL improves upon the computing time of KMS by a factor up to 27 ($d_X = 6$, $n = 1,000$). Note that both LL and KMS use local linearization, so conducting the test for a given point is

⁷For instance, for sample s , if $m = 40$ gives the largest CI, record $t_s = t_s^{(40)}$. If both $m = 20$ and $m = 40$ give the largest CI, $t_s = \min(t_s^{(20)}, t_s^{(40)})$. Here $t_s^{(m)}$ is the computing time for sample s with m being the number of initial points to use in multi-start.

fast.⁸ However, LL, as a profile-based method, only conducts the test for a grid of r , which is one dimensional, while KMS, as a calibrated projection-based method, conducts the test for a grid of θ , whose highest dimension is 15. The grid search over a relatively large dimension can be time consuming even with the E-A-M algorithm, and thus the computational advantage of LL over KMS is more prominent with larger d_θ .

Table 2: Simulation Results: Average Time

		Time (seconds per simulation)			Ratio to LL		
		$d_X = 2$	$d_X = 4$	$d_X = 6$	$d_X = 2$	$d_X = 4$	$d_X = 6$
$N = 1,000$	LL	30	37	55	1	1	1
	BCS-add	344	762	1742	11	21	31
	BCS-gen	897	2502	5037	30	68	91
	KMS	87	549	1481	3	15	27
$N = 4,000$	LL	37	50	81	1	1	1
	BCS-add	373	774	1538	10	16	19
	BCS-gen	1788	5602	13914	49	113	172
	KMS	115	362	1019	3	7	13

In Table 3, I decompose the profiling method and report the computing time per grid point with $B = 600$ without parallelization. The computing time of simple grid search using the profiling method is approximately linear in the bootstrap draws B and the number of evaluated grid points h , thus readers can estimate the computing time using simple grid search. For instance, for $n = 1000$, $d_X = 2$, 1,000 bootstrap draws, a grid search of LL over 100 points, paralleled with 4 workers will take approximately

$$(0.47 + \frac{0.13}{600} \times 1000) \times 100/4 = 17 \text{ s}.$$

The computing time of LL critical value is almost negligible compared to BCS.

Tuning parameters

One main price of LL is that the user needs to choose additional tuning parameters λ_n and set $\Delta_n^r(\theta)$. In this part, I check the sensitivity of the simulation results to these choices. I run the simulation for $n = 4000$, $d_X = 2$ with 25 different combinations of $(\lambda_n, \Delta_n^r(\theta))$ over a large range, where $\Delta_n^r(\theta)$ is given in (32), and the results are reported in Table 4. The simulation results are pretty robust to the tuning parameters and the coverage rates are higher than the nominal value for all combinations, which may be a result of the conservativeness of Test LL. The test becomes more conservative as λ_n increases. The intuition is that with larger λ_n , we penalize more the deviation from $\hat{\theta}_{n,r}$ when we calculate the bootstrapped version of $\hat{\theta}_{n,r}$, and thus we tend to search over a smaller neighborhood around $\hat{\theta}_{n,r}$, which leads to a larger critical value.

⁸Here “a given point” means a given r for LL test, or a given θ for KMS test.

Table 3: Simulation Results: Computing Time for Profiling Method

		Time (seconds per grid point)			Ratio to LL		
		$d_X = 2$	$d_X = 4$	$d_X = 6$	$d_X = 2$	$d_X = 4$	$d_X = 6$
$N = 1,000$							
Test Stat		0.47	1.20	2.47	-	-	-
Critical Val	LL	0.13	0.28	0.47	1	1	1
	BCS-add	147.22	410.42	785.53	1127	1447	1686
	BCS-gen	378.18	1229.71	2447.61	2896	4337	5254
$N = 4,000$							
Test Stat		0.88	2.55	5.24	-	-	-
Critical Val	LL	0.13	0.22	0.38	1	1	1
	BCS-add	192.46	376.35	772.65	1498	1705	2009
	BCS-gen	818.59	2828.71	6187.08	6371	12812	16087

With λ_n changing from $0.1\lambda_n^*$ to $10\lambda_n^*$, the confidence interval changes from $[-0.029, 0.713]$ to $[-0.067, 0.754]$. On the contrary, the change of ε_n has little impact. The intuition is that when $\hat{\theta}_{n,r}$ is close to the boundary of $\Theta(r)$, the shrinkage depends on the distance between $\hat{\theta}_{n,r}$ and $\Theta(r)$, rather than ε_n . When $\hat{\theta}_{n,r}$ is far away from the boundary of $\Theta(r)$, the shrinkage amount ε_n does not matter asymptotically since we only search locally around $\hat{\theta}_{n,r}$.

6 Empirical Application

In this section, I apply the test procedures to the binary entry game in Kline and Tamer (2016) (KT16). This empirical application is also revisited in Chen et al. (2018), Chesher and Rosen (2020) and KMS19 based on moment (in)equality models. The goal is to explain the entry decision in the airline industry and the main setting is summarized below.

All airlines are aggregated into two types: the low cost carrier (LCC) and others (OA). The payoff of j type firm entering market i is determined by two exogenous explanatory variables, namely *market presence* ($x_{j,i}^{\text{pres}}$) and *market size* (x_i^{size}), as well as the opponent's decision. The first one is a market- and firm- specific variable that measures the market presence of type j firm in market i . The market size is defined as the population at the endpoint city and it only depends on the market. Both exogenous variables are simplified to 1 if the value is larger than the median, and 0 otherwise. Assume that the payoff function is given by

$$A_{j,i} \left(\beta_j^{\text{cons}} + \beta_j^{\text{pres}} x_{j,i}^{\text{pres}} + \beta_j^{\text{size}} x_i^{\text{size}} + \delta_j A_{-j,i} + e_{j,i} \right)$$

where A_j is the decision of j type airline, which equals 1 if it enters and 0 otherwise; A_{-j} is the opponent's decision. β and δ are model parameters. A firm chooses to enter a market if the entering payoff is non-negative. $(e_{1,i}, e_{2,i})$ follows a joint normal distribution with correlation coefficient ρ and marginal mean and variance normalized to 0 and 1, respectively. I refer to KT16 for more details and further discussion. The exogenous variables $(x_{1,i}^{\text{pres}}, x_{2,i}^{\text{pres}}, x_i^{\text{size}})$ can take 8 values, and for each value, there are two moment equalities (for the probability of both

Table 4: Impact of tuning parameter choices

		(a) Median 95% CI					
		$\varepsilon_n = 0.1\sqrt{\ln(\kappa_n^2)/n}$	$0.5\sqrt{\ln(\kappa_n^2)/n}$	$\sqrt{\ln(\kappa_n^2)/n}$	$2\sqrt{\ln(\kappa_n^2)/n}$	$10\sqrt{\ln(\kappa_n^2)/n}$	
$\lambda_n =$	$0.1n^{\frac{1}{2}}\kappa_n^2$	[-0.029, 0.713]	[-0.028, 0.713]	[-0.029, 0.713]	[-0.029, 0.713]	[-0.029, 0.713]	
	$0.5n^{\frac{1}{2}}\kappa_n^2$	[-0.032, 0.716]	[-0.032, 0.716]	[-0.032, 0.716]	[-0.032, 0.716]	[-0.033, 0.716]	
	$n^{\frac{1}{2}}\kappa_n^2$	[-0.036, 0.721]	[-0.037, 0.721]	[-0.037, 0.721]	[-0.037, 0.721]	[-0.037, 0.721]	
	$2n^{\frac{1}{2}}\kappa_n^2$	[-0.044, 0.730]	[-0.044, 0.730]	[-0.044, 0.730]	[-0.044, 0.730]	[-0.045, 0.730]	
	$10n^{\frac{1}{2}}\kappa_n^2$	[-0.067, 0.754]	[-0.067, 0.754]	[-0.067, 0.754]	[-0.067, 0.754]	[-0.067, 0.754]	
		(b) Coverage Rate					
		$\varepsilon_n = 0.1\sqrt{\ln(\kappa_n^2)/n}$	$0.5\sqrt{\ln(\kappa_n^2)/n}$	$\sqrt{\ln(\kappa_n^2)/n}$	$2\sqrt{\ln(\kappa_n^2)/n}$	$10\sqrt{\ln(\kappa_n^2)/n}$	
		Lower Upper	Lower Upper	Lower Upper	Lower Upper	Lower Upper	
$\lambda_n =$	$0.1n^{\frac{1}{2}}\kappa_n^2$	97.8 97.5	97.8 97.5	97.8 97.5	97.7 97.6	97.7 97.5	
	$0.5n^{\frac{1}{2}}\kappa_n^2$	98.3 98.9	98.3 98.9	98.3 98.9	98.3 99.0	98.2 99.1	
	$n^{\frac{1}{2}}\kappa_n^2$	98.4 99.7	98.4 99.7	98.4 99.7	98.4 99.7	98.6 99.7	
	$2n^{\frac{1}{2}}\kappa_n^2$	99.1 99.9	99.1 99.9	99.1 99.9	99.1 99.9	99.1 99.9	
	$10n^{\frac{1}{2}}\kappa_n^2$	99.5 99.9	99.5 99.9	99.5 99.9	99.5 99.9	99.5 99.9	

entering and both not entering) and two moment inequalities (upper and lower bounds for the probability of only OA entering). This in sum gives 32 moment conditions.

The data is from the second quarter of the 2010 Airline Origin and Destination Survey, which contains $n = 7882$ markets. Let $\theta = (\beta_{\text{LCC}}^{\text{cons}}, \beta_{\text{LCC}}^{\text{size}}, \beta_{\text{LCC}}^{\text{pres}}, \beta_{\text{OA}}^{\text{cons}}, \beta_{\text{OA}}^{\text{size}}, \beta_{\text{OA}}^{\text{pres}}, \delta_{\text{LCC}}, \delta_{\text{OA}}, \rho)$ and $\Theta = \times_{i=1}^9 [\underline{\theta}_i, \bar{\theta}_i]$, where

$$\begin{aligned}\underline{\theta} &= (-8, \quad -2, \quad -2, \quad -8, \quad -2, \quad -2, \quad -4, \quad -4, \quad -0.85) \\ \bar{\theta} &= (\quad 2, \quad \quad 3, \quad 10, \quad \quad 2, \quad \quad 3, \quad 10, \quad \quad 0, \quad \quad 0, \quad 0.85)\end{aligned}$$

following from the bound used in KMS19. The bounds on ρ guarantee that Assumption 3.2 holds, and are also imposed in KMS19. I set $\kappa_n = \sqrt{\log(n)}$, $\lambda_n = \sqrt{n}\kappa_n^2$, $\epsilon_{n,j} = \frac{|\bar{\theta}_j - \underline{\theta}_j|}{2} \sqrt{\frac{\ln \kappa_n^2}{n}}$ and for j th parameter,

$$\Delta_{n,j}^r(\theta) = \{ \Delta : \Delta_j = 0, \Delta_i = [\sqrt{n}(\underline{\theta}_i - \theta_i + \epsilon_{n,i}), \sqrt{n}(\bar{\theta}_i - \theta_i - \epsilon_{n,i})], j \neq i \} \cup \{0_9\} \quad (33)$$

as in (27). All confidence intervals are calculated with E-A-M algorithms with $B = 500$ bootstrap draws. KMS is calculated with the number of initial points to use in multi start $m = 40$, which is used in the KMS19 code for the same empirical application. The 95% confidence intervals and computing time for each parameter are presented in Table 5. The moment condition is additively separable and the BCS computing time corresponds to “BCS-add” in Section 5.

The estimation results are qualitatively consistent with those in KT16 and KMS19. All bounds are informative. The market size (resp. market presence) has a significant positive effect on the payoff, and the effect is larger for OA (resp. LCC) firms. The entry of each airline significantly reduces the opponent’s payoff. LL and BCS confidence intervals are very close, which agrees with the findings in the Monte Carlo simulations. However, different from the simulation results, the confidence intervals of KMS are larger than LL and BCS, especially for $\beta_{\text{LCC}}^{\text{cons}}$, δ_1 and ρ . It is possible that the power advantage of calibrated projection and profiling methods is different under different models. The power comparison with KMS is interesting but outside the scope of this paper and I leave it for future research.

The computing time for of LL is significantly less than both BCS and KMS and it takes less than 20 minutes to get all LL confidence intervals, while it takes 2.5 hours for KMS and 7.5 hours for BCS.

7 Conclusion

This paper proposes a bootstrap test procedure for the null hypothesis $H_0 : c'\theta = r$ in moment inequality models, which I refer to as the local linearization based test. The test is profile-based, but to calculate the critical value, it only requires a local search around some preliminary estimator. An appealing feature is that the bootstrap iterates over quadratic programming problems, which significantly reduces the computing time. The new test controls asymptotic size uniformly over a large class of data generating processes and can be applied to construct confidence sets for subvectors by test inversion. Simulation experiments demonstrate good statistical and computational properties.

Table 5: Results for Empirical Application

(a) 95% confidence intervals

	LL	BCS	KMS
β_{LCC}^{cons}	[-1.11,-1.04]	[-1.23,-1.00]	[-2.06,-0.84]
β_{LCC}^{size}	[0.24, 0.35]	[0.23, 0.38]	[0.18, 0.42]
β_{LCC}^{pres}	[1.78, 1.88]	[1.78, 1.92]	[1.74, 1.95]
β_{OA}^{cons}	[0.47, 0.54]	[0.45, 0.54]	[0.40, 0.60]
β_{OA}^{size}	[0.42, 0.50]	[0.41, 0.51]	[0.33, 0.56]
β_{OA}^{pres}	[0.45, 0.52]	[0.43, 0.55]	[0.39, 0.58]
δ_{LCC}	[-1.25,-1.18]	[-1.29,-1.04]	[-1.45,-0.19]
δ_{OA}	[-1.38,-1.29]	[-1.41,-1.18]	[-1.46,-0.77]
ρ	[0.83, 0.85]	[0.80, 0.85]	[0.21, 0.85]

(b) Computing Time

	Time (seconds)			Ratio to LL	
	LL	BCS	KMS	BCS	KMS
β_{LCC}^{cons}	282	5176	2857	18	10
β_{LCC}^{size}	137	3444	435	25	3
β_{LCC}^{pres}	125	1827	327	15	3
β_{OA}^{cons}	108	1433	1963	13	18
β_{OA}^{size}	93	6886	723	74	8
β_{OA}^{pres}	78	2363	285	30	4
δ_{LCC}	101	2575	362	25	4
δ_{OA}	322	2102	956	7	3
ρ	108	1230	321	11	3
Total	1355	27036	8229	20	6

Appendix

A Proofs

A.1 Notation

Table 6: Important notation

Θ	the parameter space
$\Theta_I(F)$	the identified set
$\Theta(r)$	$\{\theta \in \Theta : c'\theta = r\}$
$\Theta_I(r, F)$	$\Theta(r) \cap \Theta_I(F)$
$\mathcal{P}$	the baseline distribution space
$\mathcal{L}$	the parameter space for (r, F) ,
$\mathcal{L}_0$	$\{(r, F) : F \in \mathcal{P}, r \in \{c'\theta : \theta \in \Theta_I(F)\}\}$
$\mathbb{R}_{[\pm\infty]}$	$[-\infty, +\infty]$
$\Sigma_F(\theta_1, \theta_2)$	$E_F \left[\left(m(W, \theta_1) - E_F[m(W, \theta_1)] \right) \left(m(W, \theta_2) - E_F[m(W, \theta_2)] \right)' \right]$
$D_F(\theta)$	$\text{diag}(\Sigma_F(\theta, \theta))$
$\Omega_F(\theta_1, \theta_2)$	$D_F(\theta_1)^{-\frac{1}{2}} \Sigma_F(\theta_1, \theta_2) D_F(\theta_2)^{-\frac{1}{2}}$
Ψ	$\{\Omega_F(\theta_1, \theta_2) : \theta_1, \theta_2 \in \Theta\}$
Ψ_1	$\{\Omega_F(\theta, \theta) : \theta \in \Theta\}$
$G_F(\theta)$	$\nabla_{\theta'} \left(E_F[D_F(\theta)^{-\frac{1}{2}} m(W, \theta)] \right)$
$\nu_n(\theta)$	$D_F(\theta)^{-\frac{1}{2}} \sqrt{n} \left(\bar{m}_n(\theta) - E_F(m(W, \theta)) \right)$
$h_{n,F}(\theta)$	$\sqrt{n} D_F(\theta)^{-\frac{1}{2}} E_F[m(W, \theta)]$
$\Gamma_{n,F}(r)$	$\left\{ (\theta, \ell) \in \Theta(r) \times \mathbb{R}^k : \ell = \sqrt{n} D_F^{-1/2}(\theta) E_F[m(W, \theta)] \right\}$

Let $\ell^\infty(\Theta)$ be the set of all uniformly bounded functions that map $\Theta \rightarrow \mathbb{R}^u$, $u \in \mathbb{N}$, equipped with the supremum norm. The sequence of probability spaces is denoted as $\{(\mathcal{W}, \mathcal{A}, F_n)\}_{n \geq 1}$. Stochastic processes W maps $\mathcal{W} \rightarrow \ell^\infty(\Theta)$. In this context, I use $\xrightarrow{d}$, $\xrightarrow{p}$, and $\xrightarrow{\text{a.s.}}$ to denote weak convergence, convergence in probability, and convergence almost surely in the $\ell^\infty(\Theta)$ metric, respectively, in the sense of [Van Der Vaart and Wellner \(1996\)](#). For any $u \in \mathbb{N}$, for any $x_1, x_2 \in \mathbb{R}_{[\pm\infty]}^u$, we define matrix

$$\tilde{d}(x_1, x_2) = \left(\sum_{i=1}^u (\vartheta(x_{1,i}) - \vartheta(x_{2,i}))^2 \right)^{1/2},$$

where $\vartheta : \mathbb{R}_{[\pm\infty]} \rightarrow [0, 1]$ is such that $\vartheta(-\infty) = 0$, $\vartheta(\infty) = 1$, and $\vartheta(y) = \Phi(y)$ for $y \in \mathbb{R}$, where Φ is the standard normal CDF. Let d_H denote the Hausdorff metric associated with $\tilde{d}$. Specifically, for any sets $A, B \in \Theta \times \mathbb{R}_{[\pm\infty]}^k$,

$$d_H(A, B) = \max \left\{ \sup_{(\theta_1, h_1) \in A} \inf_{(\theta_2, h_2) \in B} \tilde{d}((\theta_1, h_1), (\theta_2, h_2)), \right. \\ \left. \sup_{(\theta_2, h_2) \in B} \inf_{(\theta_1, h_1) \in A} \tilde{d}((\theta_1, h_1), (\theta_2, h_2)) \right\}. \quad (34)$$

I use $\xrightarrow{H}$ to denote convergence in the Hausdorff metric, that is, $A_n \xrightarrow{H} B \iff d_H(A_n, B) \rightarrow 0$. Let $\mathcal{C}(\Theta^2)$ denote the space of continuous functions that map Θ^2 to Ψ , and $\mathcal{S}(\Theta \times \mathbb{R}_{[\pm\infty]}^k)$ denote the space of compact subsets of the metric space $(\Theta \times \mathbb{R}_{[\pm\infty]}^k, \tilde{d})$. For nonstochastic functions of $\theta \in \Theta$, I use $\xrightarrow{u}$ to denote uniform in θ convergence, for example, $\Omega_{F_n} \xrightarrow{u} \Omega \iff \sup_{\theta_1, \theta_2 \in \Theta} \tilde{d}(\Omega_{F_n}(\theta_1, \theta_2), \Omega(\theta_1, \theta_2)) \rightarrow 0$. Let d denote the Euclidean distance.

I use $\Omega(\theta)$ and $\Omega(\theta, \theta)$ equivalently. And for a set Θ_1 and a scalar ϵ , the ϵ -expansion of $\Theta_1 \subset \Theta$ is defined as $\Theta_1^\epsilon := \{\theta \in \Theta : d(\theta, \Theta_1) \leq \epsilon\}$. For a vector a , $\|a\| = \sqrt{a'a}$ and for a matrix A , $\|A\| = \|\text{vec}(A)\|$. For a scalar x , $[x]_+ = \max(x, 0)$ and $[x]_- = \max(-x, 0)$. For a k dimensional vector $x = (x_1, \dots, x_k)$, let $[x]_0 = ([x_1]_+, \dots, [x_p]_+, 0, \dots, 0)$, $[x]_a = (-[x_1]_-, \dots, -[x_p]_-, x_{p+1}, \dots, x_k)$. For a set A , an element a and a scalar α , denote $A+a = \{\tilde{a}+a : \tilde{a} \in A\}$ and $\alpha A = \{\alpha a : a \in A\}$. The projection $T^r : \Theta \rightarrow \Theta$ is defined by

$$T^r \theta \in \underset{\tilde{\theta} \in \Theta_I(r, F)}{\text{argmin}} d(\tilde{\theta}, \theta).$$

For positive sequences $\{a_n\}, \{b_n\}$, I write $a_n \ll b_n$ iff $\frac{a_n}{b_n} \rightarrow 0$. The convergence rate of λ_n and κ_n are specified in Assumption 9.

A.2 Assumptions

Assumption 6. *The baseline space of distributions $\mathcal{P}$ is the set of distributions F satisfying the following properties:*

1. *The data $\{W_i\}_{i=1}^n$ are i.i.d. under F .*
2. *We have $\sigma_{F,j}^2(\theta) = \text{Var}_F(m_j(W_i, \theta)) \in (0, \infty)$ for $j = 1, \dots, k$ and all $\theta \in \Theta$.*
3. *For every $F \in \mathcal{F}$ and $j = 1, \dots, k$, $\left\{ \sigma_{F,j}^{-1}(\theta) m_j(\cdot, \theta) : \mathcal{W} \rightarrow \mathbb{R} \right\}$ is a measurable class of functions indexed by $\theta \in \Theta$.*
4. *The empirical process $\nu_n(\theta)$ with j -component*

$$\nu_{n,j}(\theta) = n^{-1/2} \sigma_{F,j}^{-1}(\theta) \sum_{i=1}^n (m_j(W_i, \theta) - E_F[m_j(W_i, \theta)]) \quad (35)$$

for $j = 1, \dots, k$ is asymptotically ρ_F -equicontinuous uniformly in $F \in \mathcal{P}$ in the sense of Van Der Vaart and Wellner (1996) page 169. This is, for any $\varepsilon > 0$

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \sup_{F \in \mathcal{P}} P_F^* \left(\sup_{\rho_F(\theta_1, \theta_2) < \delta} \|\nu_n(\theta_1) - \nu_n(\theta_2)\| > \varepsilon \right) = 0$$

where P_F^ denotes outer probability and ρ_F denotes the coordinate-wise version of the intrinsic variance semimetric defined as*

$$\rho_F(\theta_1, \theta_2) \equiv \left\| \left\{ \text{Var}_F[\sigma_{F,j}^{-1}(\theta_1) m_j(W, \theta_1) - \sigma_{F,j}^{-1}(\theta_2) m_j(W, \theta_2)]^{1/2} \right\}_{j=1}^k \right\|.$$

5. For some constant $c > 0$ and all $j = 1, \dots, k$,

$$\sup_{F \in \mathcal{F}} E_F \left[\sup_{\theta \in \Theta} \left| \frac{m_j(W, \theta)}{\sigma_{F,j}(\theta)} \right|^{2+c} \right] < \infty.$$

6. The matrix Ω_F satisfies

$$\lim_{\delta \downarrow 0} \sup_{\|(\theta_1, \tilde{\theta}_1) - (\theta_2, \tilde{\theta}_2)\| < \delta} \sup_{F \in \mathcal{P}} \left\| \Omega_F(\theta_1, \tilde{\theta}_1) - \Omega_F(\theta_2, \tilde{\theta}_2) \right\| = 0.$$

Assumption 7. For any $\{(r_n, F_n) \in \mathcal{L}_0\}_{n \geq 1}$, let (Γ, Ω) be such that $\Omega_{F_n} \xrightarrow{u} \Omega$ and $\Gamma_{n, F_n}(r_n) \xrightarrow{H} \Gamma$ with $(\Gamma, \Omega) \in \mathcal{S}(\Theta \times \mathbb{R}_{[\pm\infty]}^k) \times \mathcal{C}(\Theta^2)$ and $\Gamma_{n, F_n}(r_n)$ as in Table 6. Let $cv^S(\Gamma, \Omega, 1 - \alpha)$ be the $(1 - \alpha)$ quantile of

$$J(\Gamma, \Omega) \equiv \inf_{(\theta, \ell) \in \Gamma} S(\nu_\Omega(\theta) + \ell, \Omega(\theta)) \quad (36)$$

where $\nu_\Omega : \Theta \rightarrow \mathbb{R}^k$ is a tight Gaussian process with zero-mean and covariance (correlation) kernel Ω . Then the following statements hold:

1. If $cv^S(\Gamma, \Omega, 1 - \alpha) > 0$, the distribution of $J(\Gamma, \Omega)$ is continuous at $cv^S(\Gamma, \Omega, 1 - \alpha)$.
2. If $cv^S(\Gamma, \Omega, 1 - \alpha) = 0$, $\liminf_{n \rightarrow \infty} P_{F_n}(J_n(r_n) = 0) \geq 1 - \alpha$, where $J_n(r_n)$ is as in (4).

Assumption 8. The test function S satisfies the following properties:

1. $S((m_1, m_2), \Sigma)$ is nonincreasing in m_1 for all $m_1 \in \mathbb{R}^p$, $m_2 \in \mathbb{R}^{k-p}$, and variance matrices $\Sigma \in \mathbb{R}^{k \times k}$.
2. $S(m, \Sigma) = S(Dm, D\Sigma D)$ for all $m \in \mathbb{R}^k$, variance matrices $\Sigma \in \mathbb{R}^{k \times k}$, and pd diagonal $D \in \mathbb{R}^{k \times k}$.
3. $S(m, \Omega) \geq 0$ for all $m \in \mathbb{R}^k$ and $\Omega \in \Psi_1$.
4. $S(m, \Omega)$ is continuous at all $m \in \mathbb{R}_{[\pm\infty]}^p \times \mathbb{R}^{k-p}$ and $\Omega \in \Psi_1$.
5. For some $\chi > 0$, $S(cm, \Omega) = c^\chi S(m, \Omega)$ for all scalar $c > 0$, $m \in \mathbb{R}^k$ and $\Omega \in \Psi_1$.
6. We have $S(m, \Omega) > 0$ if and only if $m_j < 0$ for some $j = 1, \dots, p$ or $m_j \neq 0$ for some $j = p + 1, \dots, k$, where $m = (m_1, \dots, m_k)'$ and $\Omega \in \Psi_1$. Equivalently, $S(m, \Omega) = 0$ if and only if $m_j \geq 0$ for all $j = 1, \dots, p$ and $m_j = 0$ for all $j = p + 1, \dots, k$, where $m = (m_1, \dots, m_k)'$ and $\Omega \in \Psi_1$.
7. For all $n \geq 1$, $S(\sqrt{n}\bar{m}_n(\theta), \hat{\Sigma}_n(\theta))$ is a lower semicontinuous function of $\theta \in \Theta$.

Assumption 9. Rate of sequences

1. $\kappa_n \rightarrow \infty$ and $\kappa_n = O\left(\frac{n^{1/4}}{(\ln n)^{1/\chi}}\right)$.
2. $\frac{\lambda_n}{\sqrt{n}} \rightarrow \infty$ and $\lambda_n(\ln \kappa_n^2)^{\frac{2}{\chi}} = O(n)$ where χ is defined in Assumption 8.5.

A.3 Proof of Theorem 1

A.3.1 Part 1

Define $cv_n(\check{\Theta}, \check{\Omega}, \check{\varphi}, \check{G}, 1 - \alpha)$ with $\check{\Theta} \subset \Theta$, $\check{\Omega} \in \mathcal{C}(\Theta^2)$, $\check{\varphi} : \Theta \rightarrow \mathbb{R}$, $\check{G} : \Theta \rightarrow \mathbb{R}^{k \times d_\theta}$ to be the $1 - \alpha$ quantile of

$$\inf_{\theta \in \check{\Theta}} S(v(\theta) + \check{\varphi}(\theta) + \check{G}(\theta)\Delta(\theta), \check{\Omega}(\theta))$$

with v being a Gaussian process with covariance kernel $\check{\Omega}$, and

$$\Delta(\theta) \in \arg \min_{\Delta \in \Delta_n^{c'\theta}(\theta)} \left(S(v(\theta) + \check{\varphi}(\theta) + \check{G}(\theta)\Delta, \check{\Omega}(\theta)) + \frac{\lambda_n}{n} \Delta' \Delta \right).$$

Note that

$$cv_n^{\text{LL}}(r, 1 - \alpha) = cv_n(\hat{\Theta}_I(r), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha), \quad (37)$$

thus we are interested in

$$\limsup_{n \rightarrow \infty} \sup_{(r, F) \in \mathcal{L}_0} P_F \left(J_n(r) > cv_n(\hat{\Theta}_I(r), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) \right). \quad (38)$$

There is always a subsequence $\{a_n\}$, $\{r_{a_n}, F_{a_n}\}$ s.t.

$$(38) = \lim_{n \rightarrow \infty} P_{F_{a_n}} \left(J_{a_n}(r_{a_n}) > cv_{a_n}(\hat{\Theta}_I(r_{a_n}), \hat{\Omega}_{a_n}, \varphi_{a_n}, \hat{G}_{a_n}, 1 - \alpha) \right).$$

By Lemma S.3.2 in BCS17, there is a further sequence $\{u_n\}_{n \geq 1}$ of $\{a_n\}_{n \geq 1}$ s.t. $\Omega_{F_{u_n}} \xrightarrow{u} \Omega$, $\Gamma_{u_n, F_{u_n}}(r_{u_n}) \xrightarrow{H} \Gamma$, for some $(\Gamma, \Omega) \in \mathcal{S}(\Theta \times \mathbb{R}_{[\pm\infty]}^k) \times \mathcal{C}(\Theta^2)$. Here $\Gamma_{n, F}(r)$ is defined in Table 6. Then, since $\Omega_{F_{u_n}} \xrightarrow{u} \Omega$ and $\Gamma_{u_n, F_{u_n}}(r_{u_n}) \xrightarrow{H} \Gamma$, Theorem S.2.4 in BCS17 implies that

$$J_{u_n}(r_{u_n}) \xrightarrow{d} J(\Gamma, \Omega) \quad (39)$$

where $J(\Gamma, \Omega)$ is defined in (36).

A.3.2 Part 2

In this part, I show that for a sequence $(r_n, F_n) \in \mathcal{F}_0$ with $\Omega_{F_n} \xrightarrow{u} \Omega$, $\Gamma_{n, F_n}(r_n) \xrightarrow{H} \Gamma$, for some $(\Gamma, \Omega) \in \mathcal{S}(\Theta \times \mathbb{R}_{[\pm\infty]}^k) \times \mathcal{C}(\Theta^2)$, for all $\epsilon > 0$

$$\limsup_{n \rightarrow \infty} P_{F_n} \left(cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) = 0$$

where $cv^S(\Gamma, \Omega, 1 - \alpha)$ is the $1 - \alpha$ quantile of $J(\Gamma, \Omega)$ defined in (36). And it also holds if we change n to a subsequence u_n .

Let $\tau_{1n} = a\varepsilon_n$ where $\varepsilon_n = n^{-1/2} (\ln \kappa_n^2)^{\frac{1}{\chi}}$ and $a > 0$ are specified in Assumption 4. By Assumption 9.2,

$$\tau_{1n} = O \left(\lambda_n^{-\frac{1}{2}} \right). \quad (40)$$

Let τ_{2n} satisfy $\frac{\kappa_n}{\sqrt{n}} \ll \tau_{2n} \ll \sqrt{\frac{\lambda_n}{n}}$. By BCS17 Lemma S.3.1.2, we have $\hat{\Omega}_n \xrightarrow{p} \Omega$ in $l^\infty(\Theta)$. Thus

there is a sequence $\tau_{3n} = o(1)$ such that $\|\hat{\Omega}_n - \Omega\| = o_p(\tau_{3n})$. Choose τ_{4n} s.t. $\frac{(\ln \kappa_n^2)^{\frac{1}{\chi}}}{\kappa_n} \ll \tau_{4n} \ll$
1. Define a sequence of sets A_n

$$A_n = \left\{ (\check{\Theta}, \check{\Omega}, \check{\varphi}, \check{G}) : \check{\Theta} \subseteq \Theta(r_n) \cap \Theta_I(r_n, F_n)^{\tau_{1n}}, \sup_{\theta \in \check{\Theta}} \|\check{G}(\theta) - G_{F_n}(\theta)\| \leq \tau_{2n}, \right. \\ \left. \sup_{\theta_1, \theta_2 \in \check{\Theta}} \|\check{\Omega}(\theta_1, \theta_2) - \Omega_{F_n}(\theta_1, \theta_2)\| \leq \tau_{3n}, \sup_{\theta \in \check{\Theta}} \| [h_{F_n}(T^{r_n}\theta) - \check{\varphi}(\theta)]_a \| \leq \tau_{4n} \right\}. \quad (41)$$

Step 1. In this step, I consider sequences θ_n and $\tilde{\theta}_n$ such that

$$\theta_n \in \Theta(r_n) \cap \Theta_I(r_n, F_n)^{\tau_{1n}}, \quad (42)$$

$$\tilde{\theta}_n = T^{r_n}\theta_n + O_p(\lambda_n^{-\frac{1}{2}}), \quad (43)$$

and

$$\left[\nu_{F_n}(\theta_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) \right]_a = O_p(1). \quad (44)$$

I show

$$S(\hat{D}_n(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\tilde{\theta}_n), \hat{\Omega}_n(\tilde{\theta}_n)) \\ = S\left(\nu_{F_n}(\theta_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n), \Omega_{F_n}(\theta_n)\right) + o_p(1). \quad (45)$$

First, simplify $D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\tilde{\theta}_n)$, Note that

$$D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\tilde{\theta}_n) \\ = D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\left(\bar{m}_n(\tilde{\theta}_n) - E_{F_n}(m(W, \tilde{\theta}_n))\right) + \sqrt{n}D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}E_{F_n}(m(W, \tilde{\theta}_n)) \\ = \nu_{F_n}(\tilde{\theta}_n) + \sqrt{n}D_{F_n}(T^{r_n}\theta_n)^{-\frac{1}{2}}E_{F_n}(m(W, T^{r_n}\theta_n)) + G_{F_n}(\tilde{\theta}_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) \\ = \nu_{F_n}(\tilde{\theta}_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) \\ + (G_{F_n}(\tilde{\theta}_n) - G_{F_n}(\theta_n))\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) \\ = \nu_{F_n}(\theta_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) + o_p(1). \quad (46)$$

The first and third equalities are just rearranging terms; the second one is by the mean value theorem where $\bar{\theta}_n$ is between $\tilde{\theta}_n$ and $T^{r_n}\theta_n$. Note that the Taylor expansion is for k rows and $\bar{\theta}_n$ may be different for each row, but I use $\bar{\theta}_n$ for simplicity. To get the last equality, first, note that

$$\|\tilde{\theta}_n - \theta_n\| \leq \|\tilde{\theta}_n - T^{r_n}\theta_n\| + \|T^{r_n}\theta_n - \theta_n\| \\ \leq O_p(\lambda_n^{-1/2}) + \tau_{1n} \\ = O_p(\lambda_n^{-1/2}) \quad (47)$$

where the second line follows from (42) and (43), and the third one follows from (40). And thus by Assumption 6.4

$$\nu_{F_n}(\theta_n) = \nu_{F_n}(\tilde{\theta}_n) + o_p(1). \quad (48)$$

And second, by Assumption 3.2, it holds that

$$\begin{aligned}\|G_{F_n,j}(\bar{\theta}_n) - G_{F_n,j}(\theta_n)\| &\leq M_G \|\bar{\theta}_n - \theta_n\| \\ &\leq M_G \|\tilde{\theta}_n - \theta_n\| \\ &= O_p(\lambda_n^{-\frac{1}{2}}).\end{aligned}\tag{49}$$

Combine (47), (49) and Assumption 9.2, and we get

$$(G_{F_n}(\bar{\theta}_n) - G_{F_n}(\theta_n))\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) = O_p(\lambda_n^{-\frac{1}{2}})\sqrt{n}O_p(\lambda_n^{-\frac{1}{2}}) = o_p(1).\tag{50}$$

(48) and (50) imply (46).

Thus, for all $\tilde{\theta}_n = T^{r_n}\theta_n + O_p(\lambda_n^{-\frac{1}{2}})$, it holds that

$$\begin{aligned}&S(\hat{D}_n(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\tilde{\theta}_n), \hat{\Omega}_n(\tilde{\theta}_n)) \\ &= S\left(D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\tilde{\theta}_n), D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\hat{D}_n(\tilde{\theta}_n)^{\frac{1}{2}}\hat{\Omega}_n(\tilde{\theta}_n)\hat{D}_n(\tilde{\theta}_n)^{\frac{1}{2}}D_{F_n}(\tilde{\theta}_n)^{-\frac{1}{2}}\right) \\ &= S\left(\nu_{F_n}(\theta_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n) + o_p(1), \Omega_{F_n}(\theta_n) + o_p(1)\right)\end{aligned}\tag{51}$$

where the first line follows from Assumption 8.2, and the second line from (46), and

$$D_{F_n}(\theta_n)^{-\frac{1}{2}}\hat{D}_n(\theta_n)^{\frac{1}{2}} = I_k + o_p(1)$$

$$\hat{\Omega}_n(\theta_n) = \hat{\Omega}_n(\theta_n) - \Omega_{F_n}(\theta_n) + \Omega_{F_n}(\theta_n) = \Omega_{F_n}(\theta_n) + o_p(1)$$

by BCS17 Lemma S.3.1.3 and BCS17 Lemma S.3.1.2, respectively.

Lastly, by (44) and Lemma 1, we have

$$(51) = S\left(\nu_{F_n}(\theta_n) + h_{n,F_n}(T^{r_n}\theta_n) + G_{F_n}(\theta_n)\sqrt{n}(\tilde{\theta}_n - T^{r_n}\theta_n), \Omega_{F_n}(\theta_n)\right) + o_p(1),$$

which completes the proof of this step.

Step 2. Then I show that for $\{\check{\Theta}_n, \check{\Omega}_n, \check{\varphi}_n, \check{G}_n\} \in A_n$ with A_n defined in (41),

$$\liminf_{n \rightarrow \infty} cv_n(\check{\Theta}_n, \check{\Omega}_n, \check{\varphi}_n, \check{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon.\tag{52}$$

Recall that

$$\nu_n(\theta) = D_{F_n}(\theta)^{-\frac{1}{2}}\sqrt{n}\left(\bar{m}_n(\theta) - E_{F_n}(m(W, \theta))\right)$$

and by Lemma D.2.1 in BCS15, we have $\nu_n \xrightarrow{d} \nu$ where ν is a Gaussian process with covariance kernel Ω . Define v_n as a Gaussian process with covariance kernel $\check{\Omega}_n$, thus we also have $v_n \xrightarrow{d} \nu$ on $\ell^\infty(\Theta)$. By Lemma 2, there is $\tilde{v}_n \sim v_n$ s.t. $\nu_n - \tilde{v}_n \xrightarrow{p} 0$ on $\ell^\infty(\Theta)$. Define

$$\tilde{\Delta}_n(\theta) \in \underset{\Delta \in \Delta_n^{r_n}(\theta)}{\operatorname{argmin}} S\left(\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\Delta, \check{\Omega}_n(\theta)\right) + \frac{\lambda_n}{n}\Delta'\Delta.\tag{53}$$

Since $\tilde{\Delta}_n(\theta) \in \Delta_n^{r_n}(\theta)$ and by construction $\check{\Theta}_n \subset \Theta(r_n) \cap \Theta_I(r_n, F_n)^{r_{1n}}$, Assumption 4.2 implies

that for all $\theta \in \check{\Theta}_n$

$$\tilde{\Delta}_n(\theta) \in \sqrt{n}(\Theta(r_n) - T^{r_n}\theta)$$

with probability approaching 1. This is equivalent to

$$\left\{ T^{r_n}\theta + \frac{1}{\sqrt{n}}\tilde{\Delta}_n(\theta) : \theta \in \check{\Theta}_n \right\} \subset \Theta(r_n) \quad (54)$$

with probability approaching 1. Then we have

$$\begin{aligned} J_n(r_n) &= \inf_{\theta \in \Theta(r_n)} S\left(\hat{D}_n(\theta)^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(\theta), \hat{\Omega}_n(\theta)\right) \\ &\leq \inf_{\theta \in \check{\Theta}_n} S\left(\hat{D}_n(T^{r_n}\theta + \frac{1}{\sqrt{n}}\tilde{\Delta}_n(\theta))^{-\frac{1}{2}}\sqrt{n}\bar{m}_n(T^{r_n}\theta + \frac{1}{\sqrt{n}}\tilde{\Delta}_n(\theta)), \hat{\Omega}_n(T^{r_n}\theta + \frac{1}{\sqrt{n}}\tilde{\Delta}_n(\theta))\right) + o_p(1) \\ &= \inf_{\theta \in \check{\Theta}_n} S\left(\nu_n(\theta) + h_{n,F_n}(T^{r_n}\theta) + G_{F_n}(\theta)\tilde{\Delta}_n(\theta), \Omega_{F_n}(\theta)\right) + o_p(1) \end{aligned} \quad (55)$$

the first inequality follows from (54). Then I show (55). First note that $0 \in \Delta_n^{r_n}(\theta)$, thus (53) implies

$$S\left(\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)\right) + \frac{\lambda_n}{n} \left\| \tilde{\Delta}_n(\theta) \right\|^2 \leq S\left(\tilde{v}_n(\theta) + \check{\varphi}_n(\theta), \check{\Omega}_n(\theta)\right) = O_p(1).$$

Therefore, on $\ell^\infty(\check{\Theta}_n)$

$$\tilde{\Delta}_n(\theta) = O_p\left(\sqrt{\frac{n}{\lambda_n}}\right), \quad (56)$$

$$\left[\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) \right]_a = O_p(1). \quad (57)$$

Second, on $\ell^\infty(\check{\Theta}_n)$

$$\begin{aligned} &\left\| \left[\nu_n(\theta) + h_{n,F_n}(T^{r_n}\theta) + G_{F_n}(\theta)\tilde{\Delta}_n(\theta) \right]_a \right\| \\ &\leq \left\| \left[\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) \right]_a \right\| \\ &\quad + \left\| \left[(\nu_n(\theta) - \tilde{v}_n(\theta)) + (G_{F_n}(\theta) - \check{G}_n(\theta))\tilde{\Delta}_n(\theta) \right]_a \right\| + \left\| [h_{n,F_n}(T^{r_n}\theta) - \check{\varphi}_n(\theta)]_a \right\| \end{aligned} \quad (58)$$

$$\begin{aligned} &\leq \left\| \left[\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) \right]_a \right\| \\ &\quad + \left\| o_p(1) + \tau_{2n}O_p\left(\sqrt{\frac{n}{\lambda_n}}\right) + o_p(\tau_{4n}) \right\| \end{aligned} \quad (59)$$

$$= \left\| \left[\tilde{v}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) \right]_a \right\| + o_p(1). \quad (60)$$

where (58) follows from $\| [x_1 + x_2]_a \| \leq \| [x_1]_a \| + \| [x_2]_a \|$, and (59) follows from (i)

$$\nu_n(\theta) - \tilde{v}_n(\theta) = o_p(1) \quad (61)$$

by the construction of $\tilde{v}_n$, and (ii)

$$G_{F_n}(\theta) - \check{G}_n(\theta) = O(\tau_{2n}) \quad (62)$$

$$[h_{n,F_n}(T^{r_n}\theta) - \check{\varphi}_n(\theta)]_a = O(\tau_{4n}) \quad (63)$$

by the construction of A_n . Therefore, by (57) and (60), we have that on $\ell^\infty(\check{\Theta}_n)$

$$\left[\nu_n(\theta) + h_{n,F_n}(T^{r_n}\theta) + G_{F_n}(\theta)\tilde{\Delta}_n(\theta) \right]_a = O_p(1). \quad (64)$$

In sum, (56), (64) and the conclusion in step 1, (45), imply (55).

Moreover,

$$\begin{aligned} & \inf_{\theta \in \check{\Theta}_n} S(\nu_n(\theta) + h_{n,F_n}(T^{r_n}\theta) + G_{F_n}(\theta)\tilde{\Delta}_n(\theta), \Omega_{F_n}(\theta)) \\ & \quad - \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \\ & \leq \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) + [h_{n,F_n}(T^{r_n}\theta) - \check{\varphi}_n(\theta)]_a + o_p(1), \Omega_{F_n}(\theta)) \\ & \quad - \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \\ & \leq \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta) + o_p(1), \check{\Omega}_n(\theta) + o_p(1)) \\ & \quad - \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \\ & \leq \sup_{\theta \in \check{\Theta}_n} \left| S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + G_{F_n}(\theta)\tilde{\Delta}_n(\theta) + o_p(1), \check{\Omega}_n(\theta) + o_p(1)) \right. \\ & \quad \left. - S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \right| = o_p(1) \end{aligned} \quad (65)$$

where the first inequality follows from (61), (62) and the second one from (63). The last equality holds by (57) and Lemma 1. Therefore by (55) and (65), we have

$$J_n(r_n) \leq \inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) + o_p(1). \quad (66)$$

Then taking limsup on both sides of the following inequality

$$\begin{aligned} & P_{F_n} \left(\inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) \\ & \leq P_{F_n} \left(\inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) - J_n(r_n) \leq -\frac{\epsilon}{2} \right) \\ & \quad + P_{F_n} \left(J_n(r_n) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \frac{\epsilon}{2} \right) \end{aligned}$$

and using (66) and $J_n(r_n) \xrightarrow{d} J(\Gamma, \Omega)$, we get

$$\begin{aligned} & \limsup_{n \rightarrow \infty} P_{F_n} \left(\inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n(\theta) + \check{\varphi}_n(\theta) + \check{G}_n(\theta)\tilde{\Delta}_n(\theta), \check{\Omega}_n(\theta)) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) \\ & \leq P(J(\Gamma, \Omega) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \frac{\epsilon}{2}) < 1 - \alpha. \end{aligned} \quad (67)$$

Moreover, $cv_n(\check{\Theta}_n, \check{\Omega}_n, \check{\varphi}_n, \check{G}_n, 1 - \alpha)$ is the $1 - \alpha$ quantile of $\inf_{\theta \in \check{\Theta}_n} S(\tilde{\nu}_n + \check{\varphi}_n + \check{G}_n\tilde{\Delta}_n, \check{\Omega}_n)$, thus

(67) implies

$$\liminf_{n \rightarrow \infty} cv_n(\check{\Theta}_n, \check{\Omega}_n, \check{\varphi}_n, \check{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon.$$

Step 3. I next show that

$$\liminf_{n \rightarrow \infty} P_{F_n} \left(cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) = 1. \quad (68)$$

We can show that

$$\liminf_{n \rightarrow \infty} P_{F_n}(\mathcal{W}_n^*) = 1. \quad (69)$$

where

$$\mathcal{W}_n^* = \left\{ (\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n) \subset A_n \right\}$$

and A_n is defined in (41). (i) Let

$$\Theta_n \equiv \left\{ \theta \in \Theta : S \left(\sqrt{n} E_F[m(W, \theta)], \Sigma_F(\theta) \right) \leq a^\chi \ln \kappa_n^2 \right\}. \quad (70)$$

Then by Lemma D.13 in BCS15, we have

$$\liminf_{n \rightarrow \infty} \inf_{(r, F) \in \mathcal{L}_0} P_F \left(\hat{\theta}_{n,r} \subset \Theta_n \cap \Theta(r) \right) = 1. \quad (71)$$

And by Assumption 2, Assumption 8.5, for all $\theta \in \Theta_n \cap \Theta(r)$, it holds that

$$d(\theta, \Theta_I(r, F)) \leq a \frac{(\ln \kappa_n^2)^{1/\chi}}{\sqrt{n}} = \tau_{1n} \quad (72)$$

when n is large enough. Thus we can rewrite (71) as

$$\liminf_{n \rightarrow \infty} \inf_{(r, F) \in \mathcal{L}_0} P_F \left(d(\hat{\theta}_{n,r}, \Theta_I(r, F)) \leq \tau_{1n} \right) = 1. \quad (73)$$

(ii) by Assumption 3.1 and BCS17 Lemma S.3.1.2,

$$\liminf_{n \rightarrow \infty} P_{F_n} \left(\sup_{\theta \in \hat{\Theta}_I(r_n)} \|\hat{G}_n(\theta) - G_{F_n}(\theta)\| \leq \tau_{2n}, \sup_{\theta \in \hat{\Theta}_I(r_n)} \|\hat{\Omega}_n(\theta) - \Omega_{F_n}(\theta)\| \leq \tau_{3n} \right) = 1$$

(iii) Note that for $\theta_n \in \hat{\Theta}_I(r_n)$

$$\begin{aligned} \varphi_n(\theta_n) &= \left[\hat{D}_n(\theta_n)^{-\frac{1}{2}} D_{F_n}(\theta_n)^{\frac{1}{2}} D_{F_n}(\theta_n)^{-\frac{1}{2}} \sqrt{n} \bar{m}_n(\theta_n) / \kappa_n \right]_0 \\ &= \left[(I_k + o_p(1)) (\nu_{F_n}(\theta_n) + h_{n, F_n}(T^{r_n} \theta_n) \right. \\ &\quad \left. + G_{F_n}(\theta_n) \sqrt{n} (\theta_n - T^{r_n} \theta_n) + o_p(1)) / \kappa_n \right]_0 \\ &= \left[(I_k + o_p(1)) h_{n, F_n}(T^{r_n} \theta_n) / \kappa_n + O_p \left((\ln \kappa_n^2)^{1/\chi} / \kappa_n \right) \right]_0 \\ &\leq h_{F_n}(T^{r_n} \theta_n) + O_p \left((\ln \kappa_n^2)^{1/\chi} / \kappa_n \right) \end{aligned} \quad (74)$$

where the first line is the definition of φ_n , the second line follows from (46), the third line from

$d(\theta_n, T^{r_n}\theta_n) = o(\tau_{1n})$ and Assumption 3.2, the last line follows from $\kappa_n \rightarrow +\infty$. Thus

$$\liminf_{n \rightarrow \infty} P_{F_n} \left(\inf_{\theta \in \hat{\Theta}_I(r_n)} \left\| \lfloor h_{F_n}(T^{r_n}\theta) - \varphi_n(\theta) \rfloor_a \right\| \leq \tau_{4n} \right) = 1.$$

By (i), (ii), (iii) above, we complete the proof of (69).

Next, note that

$$\begin{aligned} & \liminf_{n \rightarrow \infty} P_{F_n} \left(cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) \\ & \geq \liminf_{n \rightarrow \infty} P_{F_n} \left(cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \middle| \mathcal{W}_n^* \right) P_{F_n}(\mathcal{W}_n^*) \\ & \geq \liminf_{n \rightarrow \infty} P_{F_n} \left(cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \middle| \mathcal{W}_n^* \right) \liminf_{n \rightarrow \infty} P_{F_n}(\mathcal{W}_n^*) \\ & \geq \liminf_{n \rightarrow \infty} P_{F_n} \left(\inf_{(\Theta, \Omega, \varphi, G) \in A_n} cv_n(\Theta, \Omega, \varphi, G, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) \\ & \geq \liminf_{n \rightarrow \infty} P_{F_n} \left(cv_n(\Theta_n^*, \Omega_n^*, \varphi_n^*, G_n^*, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \frac{\epsilon}{2} \right) \\ & \geq P \left(\liminf_{n \rightarrow \infty} \left\{ cv_n(\Theta_n^*, \Omega_n^*, \varphi_n^*, G_n^*, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \frac{\epsilon}{2} \right\} \right) \\ & \geq P \left(\liminf_{n \rightarrow \infty} cv_n(\Theta_n^*, \Omega_n^*, \varphi_n^*, G_n^*, 1 - \alpha) > cv^S(\Gamma, \Omega, 1 - \alpha) - \frac{\epsilon}{2} \right) = 1 \end{aligned}$$

the first inequality follows from $P(\cdot) \geq P(\cdot \cap \mathcal{W}_n^*) = P(\cdot | \mathcal{W}_n^*) P(\mathcal{W}_n^*)$; the second one is straightforward; the third one follows from conditional on $\mathcal{W}_n^*$,

$$cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) \geq \inf_{(\Theta, \Omega, \varphi, G) \in A_n} cv_n(\Theta, \Omega, \varphi, G, 1 - \alpha)$$

and (69); in the fourth one, $(\Theta_n^*, \Omega_n^*, \varphi_n^*, G_n^*) \in A_n$ satisfies

$$\inf_{(\Theta, \Omega, \varphi, G) \in A_n} cv_n(\Theta, \Omega, \varphi, G, 1 - \alpha) + \frac{\epsilon}{2} > cv_n(\Theta_n^*, \Omega_n^*, \varphi_n^*, G_n^*, 1 - \alpha)$$

the fifth one follows from Fatou's lemma, the sixth one is straightforward, and the last one follows from that (52) holds for all $\epsilon > 0$.

A.3.3 Part 3

Consider $(r_n, F_n) \in \mathcal{F}_0$ with $\Omega_{F_n} \xrightarrow{u} \Omega$, $\Gamma_{n, F_n}(r_n) \xrightarrow{H} \Gamma$, where $(\Gamma, \Omega) \in \mathcal{S}(\Theta \times \mathbb{R}_{[\pm\infty]}^k) \times \mathcal{C}(\Theta^2)$.

Step 1. I first show

$$\lim_{n \rightarrow \infty} P_{F_n} \left(J_n(r_n) > cv_n(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha) + \eta \right) \leq \alpha$$

for all $\eta > 0$. Define $\epsilon > 0$ so that $\eta - \epsilon > 0$ and $cv^S(\Gamma, \Omega, 1 - \alpha) + \eta - \epsilon$ is a continuity point of the CDF of $J(\Gamma, \Omega)$. It then follows that

$$\begin{aligned} & P_{F_n} \left(J_n(r_n) > cv_n \left(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha \right) + \eta \right) \\ & \leq P_{F_n} \left(cv_n \left(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha \right) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon \right) \\ & \quad + P_{F_n} \left(J_n(r_n) > cv^S(\Gamma, \Omega, 1 - \alpha) + \eta - \epsilon \right). \end{aligned}$$

Take the limit supremum on both sides and by (68) and $J_n(r_n) \xrightarrow{d} J(\Gamma, \Omega)$, we get

$$\begin{aligned} & \limsup_{n \rightarrow \infty} P_{F_n} \left(J_n(r_n) > cv_n \left(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha \right) + \eta \right) \\ & \leq 1 - P \left(J(\Gamma, \Omega) \leq cv^S(\Gamma, \Omega, 1 - \alpha) + \eta - \epsilon \right) \leq \alpha. \end{aligned} \tag{75}$$

Step 2. For $\eta = 0$, there are two cases. First, suppose $cv^S(\Gamma, \Omega, 1 - \alpha) = 0$, then by Assumption 7,

$$\limsup_{n \rightarrow \infty} P_{F_n} \left(J_n(r_n) > cv_n \left(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha \right) \right) \leq \limsup_{n \rightarrow \infty} P_{F_n} (J_n(r_n) > 0) \leq \alpha.$$

Second, suppose $cv^S(\Gamma, \Omega, 1 - \alpha) > 0$. Consider a sequence $\{\epsilon_m\}_{m \geq 1}$ s.t. $\epsilon_m \downarrow 0$ and $cv(\Gamma, \Omega, 1 - \alpha) - \epsilon_m$ is a continuity point of the CDF of $J(\Gamma, \Omega)$ for all $m \in \mathbb{N}$. For any $m \in \mathbb{N}$, it follows from (75) that

$$\begin{aligned} & \limsup_{n \rightarrow \infty} P_{F_n} \left(J_n(r_n) > cv_n \left(\hat{\Theta}_I(r_n), \hat{\Omega}_n, \varphi_n, \hat{G}_n, 1 - \alpha \right) \right) \\ & \leq 1 - P \left(J(\Gamma, \Omega) \leq cv^S(\Gamma, \Omega, 1 - \alpha) - \epsilon_m \right). \end{aligned}$$

Taking $\epsilon_m \downarrow 0$ and using Assumption 7.1 gives the RHS equal to α . Together with the discussion in Section A.3.1 and (37), this completes the proof.

B Lemmas

Lemma 1. *If $\nu_{1n} - \nu_{2n} = o_p(1)$, $[\nu_{2n}]_a = O_p(1)$, and $\Omega_{1n}, \Omega_{2n} \in \Psi_1$, $\Omega_{1n} - \Omega_{2n} = o_p(1)$, it holds that*

$$S(\nu_{1n}, \Omega_{1n}) = S(\nu_{2n}, \Omega_{2n}) + o_p(1).$$

Proof. We want to show that for all $\epsilon > 0$, there is N_1 such that for all $n > N_1$,

$$P(|S(\nu_{1n}, \Omega_{1n}) - S(\nu_{2n}, \Omega_{2n})| > \epsilon) \leq \epsilon.$$

Since $[\nu_{2n}]_a = O_p(1)$, for all $\epsilon > 0$, there is M such that for all $n \in \mathbb{N}$,

$$P \left(\nu_{2n} \in [-M, +\infty)^p \times [-M, M]^{k-p} \right) \geq 1 - \frac{\epsilon}{3}.$$

And by Lemma 3, there is $\delta > 0$ such that $\|\nu_{1n} - \nu_{2n}\| \leq \delta$, $\|\Omega_{1n} - \Omega_{2n}\| \leq \delta$ and $\nu_{2n} \in$

$[-M, +\infty)^p \times [-M, M]^{k-p}$ imply that

$$|S(\nu_{1n}, \Omega_{1n}) - S(\nu_{2n}, \Omega_{2n})| \leq \epsilon.$$

Since $\nu_{1n} - \nu_{2n} = o_p(1)$ and $\Omega_{1n} - \Omega_{2n} = o_p(1)$, there is N_2 such that $\forall n \geq N_2$

$$P(\|\nu_{1n} - \nu_{2n}\| > \delta) \leq \frac{\epsilon}{3}, \quad P(\|\Omega_{1n} - \Omega_{2n}\| > \delta) \leq \frac{\epsilon}{3}.$$

Therefore, for $n \geq \max\{N_1, N_2\}$,

$$\begin{aligned} & P(|S(\nu_{1n}, \Omega_{1n}) - S(\nu_{2n}, \Omega_{2n})| > \epsilon) \\ & \leq 1 - P\left(\nu_{2n} \in [-M, +\infty)^p \times [-M, M]^{k-p}, \|\nu_{1n} - \nu_{2n}\| \leq \delta, \|\Omega_{1n} - \Omega_{2n}\| \leq \delta\right) \\ & \leq P\left(\nu_{2n} \notin [-M, +\infty)^p \times [-M, M]^{k-p}\right) + P(\|\nu_{1n} - \nu_{2n}\| > \delta) + P(\|\Omega_{1n} - \Omega_{2n}\| > \delta) \\ & \leq \epsilon. \end{aligned}$$

□

Lemma 2. Assume that $\nu_n, v_n, \nu : \mathcal{W} \rightarrow \ell(\Theta)$, $\nu_n \xrightarrow{d} \nu$ and $v_n \xrightarrow{d} \nu$ where ν is a Gaussian process with covariance kernel $\Omega \in \mathcal{C}(\Theta^2)$, then there is $v_n^* : \mathcal{W} \rightarrow \ell(\Theta)$ (suitably enriched if needed) such that $v_n^* \sim v_n$ and $\nu_n - v_n^* \xrightarrow{p} 0$.

Proof. I first show that there are identically distributed copies of ν_n and v_n , denoted as $\tilde{\nu}_n$ and $\tilde{v}_n$, defined on $(\mathcal{W}, \mathcal{A}, F_n)$ (suitably enriched if needed) such that $\tilde{\nu}_n - \tilde{v}_n \xrightarrow{p} 0$. By almost sure representation, there is $\bar{\nu}_n \sim \nu_n$, $\bar{\nu} \sim \nu$ s.t. $\bar{\nu}_n - \bar{\nu} \xrightarrow{\text{a.s.}} 0$ and $\bar{v}_n \sim v_n$, $\bar{v} \sim \nu$ s.t. $\bar{v}_n - \bar{v} \xrightarrow{\text{a.s.}} 0$. Let α be the law of $(\bar{\nu}_n, \bar{\nu})$ and β be the law of $(\bar{v}_n, \bar{v})$. By Theorem 1.31 in [Dudley \(2014\)](#), consider $X, Y, Z = R^{d_\nu}$, $\bar{\nu}_n$ defined on X , $\bar{\nu}, \bar{v}$ defined on Y , $\bar{v}_n$ defined on Z . Since the marginal distribution of α and β on Y is the same, we can find a law γ on $X \times Y \times Z$ (denote as the law of $(\tilde{\nu}_n, \tilde{\nu}, \tilde{v}_n)$) such that $(\tilde{\nu}_n, \tilde{\nu})$ has the same law as α and $(\tilde{v}_n, \tilde{v})$ has the same law as β . Thus

$$\tilde{\nu}_n - \tilde{\nu} \xrightarrow{\text{a.s.}} 0, \tilde{v}_n - \tilde{v} \xrightarrow{\text{a.s.}} 0 \Rightarrow \tilde{\nu}_n - \tilde{v}_n \xrightarrow{\text{a.s.}} 0 \Rightarrow \tilde{\nu}_n - \tilde{v}_n \xrightarrow{p} 0.$$

Then by the same argument in [Chernozhukov et al. \(2013\)](#) Lemma 9 (change δ_n in Lemma 9 to 1), there is an identical copy v_n^* of v_n such that $\nu_n - v_n^* \xrightarrow{p} 0$. □

Lemma 3. For all $\epsilon > 0$, $M > 0$, there is $\delta > 0$ such that $\nu_1 \in [-M, \infty)^p \times [-M, M]^{k-p}$, $\|\nu_1 - \nu_2\| \leq \delta$ and $\Omega_1, \Omega_2 \in \Psi_1$, $\|\Omega_1 - \Omega_2\| \leq \delta$ imply that $|S(\nu_1, \Omega_1) - S(\nu_2, \Omega_2)| \leq \epsilon$.

Proof. First note that $\forall \nu_{11} \in [-M, \infty]^p, \nu_{12} \in [-M, M]^{k-p}$,

$$\begin{aligned} S((\nu_{11}, \nu_{12}), \Omega_1) & \leq S((-M\mathbf{1}_p, \nu_{12}), \Omega_1) \\ & \leq \max_{\tilde{\nu}_{12} \in [-M, M]^{k-p}, \Omega_1 \in \Psi_1} S((-M\mathbf{1}_p, \tilde{\nu}_{12}), \Omega_1) \\ & < \infty \end{aligned}$$

where $\mathbf{1}_p = \{ \underbrace{1, \dots, 1}_p \}$, the first inequality follows from Assumption 8.1, the second one is

straightforward, and the last one from Assumption 8.4 and the compactness of $[-M, M]^{k-p}$ and Ψ_1 . Then consider function $\bar{S} : [\Phi(-M), 1]^p \times [\Phi(-M), \Phi(M)]^{k-p} \rightarrow \mathbb{R}$ defined by

$$\bar{S}(\bar{\nu}, \Omega) = S(\Phi^{-1}(\bar{\nu}), \Omega),$$

where for $\bar{\nu} = (\bar{\nu}_1, \dots, \bar{\nu}_k) \in \mathbb{R}^k$, $\Phi^{-1}(\bar{\nu}) = (\Phi^{-1}(\bar{\nu}_1), \dots, \Phi^{-1}(\bar{\nu}_k))$. Φ is the CDF of standard Normal distribution. By Assumption 8.4, we have that $\bar{S}$ is continuous for all $\bar{\nu} \in [\Phi(-M), 1]^p \times [\Phi(-M), \Phi(M)]^{k-p}$, which implies that $\bar{S}$ is uniformly continuous on $[\Phi(-M), 1]^p \times [\Phi(-M), \Phi(M)]^{k-p}$. That is, there is $\delta > 0$ s.t. if $\|\bar{\nu}_1 - \bar{\nu}_2\| \leq \delta$ and $\|\Omega_1 - \Omega_2\| \leq \delta$, it holds that

$$|\bar{S}(\bar{\nu}_1, \Omega_1) - \bar{S}(\bar{\nu}_2, \Omega_2)| \leq \epsilon.$$

Note that $\|\nu_1 - \nu_2\| \leq \delta$ implies $\|\Phi(\nu_1) - \Phi(\nu_2)\| \leq \Phi'(0) \|\nu_1 - \nu_2\| < \delta$, thus if $\|\nu_1 - \nu_2\| \leq \delta$ and $\|\Omega_1 - \Omega_2\| \leq \delta$, it holds that

$$|S(\nu_1, \Omega_1) - S(\nu_2, \Omega_2)| = |\bar{S}(\Phi(\nu_1), \Omega_1) - \bar{S}(\Phi(\nu_2), \Omega_2)| \leq \epsilon.$$

□

C Computational details of Example 2

Let $Z_{jn} = \frac{\sqrt{n}(\bar{W}_j - E_{F_n}[W_j])}{\hat{\sigma}_j}$, $\hat{\mu}_1 = \frac{\mu}{\hat{\sigma}_1}$ and $\hat{\mu}_2 = \frac{\mu}{\hat{\sigma}_2}$. Then

$$J_n(1) = \min_{\theta_2 \in [-2, 2]} \left[Z_{1n} - \hat{\mu}_1 + \frac{\sqrt{n}(1 + \theta_2)}{\hat{\sigma}_1} \right]_-^2 + \left[Z_{2n} - \hat{\mu}_2 + \frac{\sqrt{n}(1 - \theta_2)}{\hat{\sigma}_2} \right]_-^2 + \left(Z_{3n} - \eta \frac{\sqrt{n}(1 - \theta_2)(\theta_2 + 1)}{\hat{\sigma}_3} \right)^2.$$

If θ_2 close to -1 , asymptotically only the first and third elements matter and the local minimum is

$$J_n^{\text{local},1}(1) = \frac{[2\eta(Z_{1n} - \hat{\mu}_1) + Z_{3n}]_-^2}{4\eta^2 + 1} + o_p(1)$$

with minimizer

$$\frac{\sqrt{n}(1 + \theta_2^{\text{local},1})}{\hat{\sigma}_1} = \begin{cases} \frac{2\eta Z_{3n} - Z_{1n} + \hat{\mu}_1}{1 + 4\eta^2} + o_p(1) & \text{if } 2\eta(Z_{1n} - \hat{\mu}_1) + Z_{3n} \leq 0 \\ \frac{Z_{3n}}{2\eta} + o_p(1) & \text{if } 2\eta(Z_{1n} - \hat{\mu}_1) + Z_{3n} > 0 \end{cases}$$

If θ_2 close to 1 , asymptotically only the second and third elements matter and the local minimum is

$$J_n^{\text{local},2}(1) = \frac{[2\eta(Z_{2n} - \hat{\mu}_2) + Z_{3n}]_-^2}{4\eta^2 + 1} + o_p(1)$$

with minimizer

$$\frac{\sqrt{n}(1 - \theta_2^{\text{local},2})}{\hat{\sigma}_2} = \begin{cases} \frac{2\eta Z_{3n} - Z_{2n} + \hat{\mu}_2}{1 + 4\eta^2} + o_p(1) & \text{if } 2\eta(Z_{2n} - \hat{\mu}_2) + Z_{3n} \leq 0, \\ \frac{Z_{3n}}{2\eta} + o_p(1) & \text{if } 2\eta(Z_{2n} - \hat{\mu}_2) + Z_{3n} > 0. \end{cases}$$

In sum,

$$J_n(1) = \min_{j=1,2} \frac{[2\eta(Z_{jn} - \hat{\mu}_1) + Z_{3n}]_-^2}{4\eta^2 + 1}$$

with

$$\hat{\Theta}_I(1) = \begin{cases} \left\{ \left(1, \theta_2^{\text{local},1}\right) \right\} & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} < 0, Z_{1n} - \hat{\mu}_1 > Z_{2n} - \hat{\mu}_2 \\ \left\{ \left(1, \theta_2^{\text{local},2}\right) \right\} & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} < 0, Z_{1n} - \hat{\mu}_1 < Z_{2n} - \hat{\mu}_2 \\ \left\{ \left(1, \theta_2^{\text{local},1}\right), \left(1, \theta_2^{\text{local},2}\right) \right\} & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} \geq 0. \end{cases} \quad (76)$$

Note that (76) omits some $o_p(1)$ term to simplify the notation. And it happens with probability approaching zero that $J_n(1) > 0$ and $\hat{\Theta}_I(1)$ contains more than one element. In sum, it holds that

$$J_n(1) \xrightarrow{d} \min_{j=1,2} \frac{[2\eta(Z_j - \mu) + Z_3]_-^2}{4\eta^2 + 1},$$

where $Z = (Z_1, Z_2, Z_3) \sim N(0, I_3)$.

Test PR. We have

$$J_n^{\text{PR}}(1) = \min_{\theta_2 \in [-2, 2]} \left[v_{n,1}^* + \frac{\sqrt{n}(1 + \theta_2)}{\kappa_n \hat{\sigma}_1} \right]_-^2 + \left[v_{n,2}^* + \frac{\sqrt{n}(1 - \theta_2)}{\kappa_n \hat{\sigma}_1} \right]_-^2 + \left(v_{n,3}^* - \frac{\sqrt{n}\eta(1 - \theta_2^2)}{\kappa_n \hat{\sigma}_3} \right) + o_p(1)$$

Similarly, the local minimizer $\hat{\theta}_2^{\text{PR}}$ is close to either -1 or 1 . If $\hat{\theta}_2^{\text{PR}}$ close to -1 , the local minimum is

$$J_n^{\text{PR,local},1}(1) = \frac{[2\eta v_{n,1}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1},$$

at

$$\frac{\sqrt{n}(1 + \theta_2^{\text{PR,local},1})}{\kappa_n \hat{\sigma}_1} = \begin{cases} \frac{2\eta v_{n,3}^* - v_{n,1}^*}{1 + 4\eta^2} & \text{if } 2\eta v_{n,1}^* + v_{n,3}^* \leq 0, \\ \frac{v_{n,3}^*}{2\eta} & \text{if } 2\eta v_{n,1}^* + v_{n,3}^* > 0. \end{cases}$$

If θ_2 close to 1 , the local minimum is

$$J_n^{\text{PR,local},2}(1) = \frac{[2\eta v_{n,2}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1},$$

at

$$\frac{\sqrt{n}(1 - \theta_2^{\text{PR,local},2})}{\kappa_n \hat{\sigma}_2} = \begin{cases} \frac{2\eta v_{n,3}^* - v_{n,2}^*}{1 + 4\eta^2} & \text{if } 2\eta v_{n,2}^* + v_{n,3}^* \leq 0, \\ \frac{v_{n,3}^*}{2\eta} & \text{if } 2\eta v_{n,2}^* + v_{n,3}^* > 0. \end{cases}$$

In sum,

$$J_n^{\text{PR}}(1) = \min_{j=1,2} \frac{[2\eta v_{n,j}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1}.$$

Thus we can show that

$$J_n^{\text{PR}}(1) \xrightarrow{d} \min_{j=1,2} \frac{[2\eta Z_j + Z_3]_-^2}{4\eta^2 + 1}.$$

Test LL The bootstrapped test statistic $J_n^{\text{LL}}(1)$ depends on $\hat{\Theta}_I(1)$. If $\hat{\Theta}_I(1)$ is singleton with

$\hat{\theta}_2$ close to -1 , we have

$$J_n^{\text{LL}}(1) = \left[v_{n,1}^* + \frac{\Delta_2^{\text{LL}}}{\hat{\sigma}_1} \right]_-^2 + \left(v_{n,3}^* - 2\eta \frac{\Delta_2^{\text{LL}}}{\hat{\sigma}_3} \right)^2 + o_p(1)$$

where

$$\Delta_2^{\text{LL}} \in \arg \min_{\Delta \in \sqrt{n}[-2-\hat{\theta}_2+\varepsilon_n, 2-\hat{\theta}_2-\varepsilon_n]} \left[v_{n,1}^* + \frac{\Delta}{\hat{\sigma}_1} \right]_-^2 + \left(v_{n,3}^* - 2\eta \frac{\Delta}{\hat{\sigma}_3} \right)^2 + \frac{\lambda_n}{n} \Delta^2.$$

This simplifies to

$$J_n^{\text{LL}}(1) = \frac{[2\eta v_{n,1}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1} + o_p(1).$$

Similarly, If $\hat{\Theta}_I(r)$ is singleton with $\hat{\theta}_2$ close to 1, we have

$$J_n^{\text{LL}}(1) = \frac{[2\eta v_{n,2}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1} + o_p(1).$$

And if $\hat{\Theta}_I(r)$ contains more than one element, we have

$$J_n^{\text{LL}}(1) = \min_{j=1,2} \frac{[2\eta v_{n,j}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1} + o_p(1).$$

Let δ to be an indicator for the type of $\hat{\Theta}_I(1)$, i.e.

$$\delta = \begin{cases} 1 & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} < 0, Z_{1n} - \hat{\mu}_1 > Z_{2n} - \hat{\mu}_2 \\ 2 & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} < 0, Z_{1n} - \hat{\mu}_1 < Z_{2n} - \hat{\mu}_2 \\ 3 & \text{if } 2\eta(\min_j Z_{jn} - \hat{\mu}_j) + Z_{3n} \geq 0 \end{cases}$$

In sum

$$J_n^{\text{LL}}(1) = \mathbf{1}[\delta = 1] \frac{[2\eta v_{n,1}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1} + \mathbf{1}[\delta = 2] \frac{[2\eta v_{n,2}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1} + \mathbf{1}[\delta = 3] \min_{j=1,2} \frac{[2\eta v_{n,j}^* + v_{n,3}^*]_-^2}{4\eta^2 + 1}.$$

D Additional simulation results

In this section, I report the main simulation results of parameter ρ with the same setting and implementation details in Section 5. I increase the sample sizes to $N = 4,000$ and $N = 8,000$ because the median CI is not informative, i.e. $[0, 0.85]$, for all three methods with $N = 1,000$.

Confidence intervals and coverage rates

The median 95% CI and coverage rate are reported in Table 7. All upper bounds of the CIs are 0.85, which is the maximum feasible value. The lower bounds of LL and BCS are similar, while the ones of KMS are strictly higher. All coverage rates are higher than the nominal value.

Table 7: Simulation Results: Median CI and Coverage

		(a) Median 95% CI					
		$d_X = 2$		$d_X = 4$		$d_X = 6$	
Identified Set		[0.362, 0.493]		[0.362, 0.471]		[0.364, 0.471]	
$N = 4,000$	LL	[0.069, 0.85]		[0.043, 0.85]		[0.006, 0.85]	
	BCS	[0.054, 0.85]		[0.043, 0.85]		[0.023, 0.85]	
	KMS	[0.098, 0.85]		[0.086, 0.85]		[0.066, 0.85]	
$N = 8,000$	LL	[0.100, 0.85]		[0.083, 0.85]		[0.062, 0.85]	
	BCS	[0.089, 0.85]		[0.081, 0.85]		[0.069, 0.85]	
	KMS	[0.117, 0.85]		[0.112, 0.85]		[0.099, 0.85]	

		(b) Coverage Rate					
		$d_X = 2$		$d_X = 4$		$d_X = 6$	
		Lower	Upper	Lower	Upper	Lower	Upper
$N = 4,000$	LL	99.8	100.0	99.6	100.0	99.6	100.0
	BCS	99.8	99.7	99.7	100.0	99.0	99.9
	KMS	98.7	98.2	97.8	98.7	97.7	98.8
$N = 8,000$	LL	98.7	99.6	98.9	100.0	99.8	100.0
	BCS	99.8	99.8	99.1	99.9	99.7	100.0
	KMS	98.0	97.6	96.7	98.9	97.7	98.8

Computing time

The average computing time is reported in Table 8. LL improves upon BCS-add by a factor ranging from 8 to 19. LL and KMS perform similarly, but there is still a computational advantage of LL over KMS with large d_θ . I omit the computing time of BCS-gen as it becomes infeasible with large sample sizes.

Table 8: Simulation Results: Average Time

		Time (seconds per simulation)			Ratio to LL		
		$d_X = 2$	$d_X = 4$	$d_X = 6$	$d_X = 2$	$d_X = 4$	$d_X = 6$
$N = 4,000$	LL	28	39	65	1	1	1
	BCS-add	392	586	1228	13.8	14.9	18.9
	KMS	29	79	147	1.0	2.0	2.2
$N = 8,000$	LL	40	51	146	1	1	1
	BCS-add	506	487	1153	12.6	9.5	7.9
	KMS	31	66	172	0.8	1.3	1.2

Tuning parameters

I run the simulation for $n = 4000$, $d_X = 2$ with 25 different combinations of $(\lambda_n, \Delta_n^r(\theta))$, where $\Delta_n^r(\theta)$ is given in (32). The results are reported in Table 9. The length of the confidence interval slightly increases in λ_n , but does not change materially. ε_n has little impact on the results.

Table 9: Impact of tuning parameter choices

		(a) Median 95% CI					
		$\varepsilon_n = 0.1\sqrt{\ln(\kappa_n^2)/n}$		$0.5\sqrt{\ln(\kappa_n^2)/n}$		$\sqrt{\ln(\kappa_n^2)/n}$	
$\lambda_n =$	$0.1n^{\frac{1}{2}}\kappa_n^2$	[0.080, 0.85]	[0.080, 0.85]	[0.080, 0.85]	[0.080, 0.85]	$2\sqrt{\ln(\kappa_n^2)/n}$	$10\sqrt{\ln(\kappa_n^2)/n}$
	$0.5n^{\frac{1}{2}}\kappa_n^2$	[0.075, 0.85]	[0.075, 0.85]	[0.075, 0.85]	[0.075, 0.85]	[0.075, 0.85]	[0.075, 0.85]
	$n^{\frac{1}{2}}\kappa_n^2$	[0.069, 0.85]	[0.069, 0.85]	[0.069, 0.85]	[0.069, 0.85]	[0.069, 0.85]	[0.069, 0.85]
	$2n^{\frac{1}{2}}\kappa_n^2$	[0.058, 0.85]	[0.058, 0.85]	[0.058, 0.85]	[0.058, 0.85]	[0.058, 0.85]	[0.058, 0.85]
	$10n^{\frac{1}{2}}\kappa_n^2$	[0.028, 0.85]	[0.028, 0.85]	[0.028, 0.85]	[0.028, 0.85]	[0.028, 0.85]	[0.028, 0.85]
		(b) Coverage Rate					
		$\varepsilon_n = 0.1\sqrt{\ln(\kappa_n^2)/n}$		$0.5\sqrt{\ln(\kappa_n^2)/n}$		$\sqrt{\ln(\kappa_n^2)/n}$	
$\lambda_n =$	$0.1n^{\frac{1}{2}}\kappa_n^2$	Lower 99.3	Upper 98.5	Lower 99.3	Upper 98.5	Lower 99.3	Upper 99.2
	$0.5n^{\frac{1}{2}}\kappa_n^2$	99.5	99.7	99.5	99.7	99.5	99.9
	$n^{\frac{1}{2}}\kappa_n^2$	99.8	100.0	99.8	100.0	99.8	100.0
	$2n^{\frac{1}{2}}\kappa_n^2$	99.8	100.0	99.8	100.0	99.8	100.0
	$10n^{\frac{1}{2}}\kappa_n^2$	100.0	100.0	100.0	100.0	100.0	100.0

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